FAKULTA APLIKOVANÝCH VĚD
ZÁPADOČESKÉ UNIVERZITY
V PLZNI

Edgebreaker

Komprese trojúhelníkových sítí

Jakub Vokoun

Fakulta aplikovaných věd ZČU

javok@students.zcu.cz

2026-11-04

1. Motivace

1.1 Dobový kontext

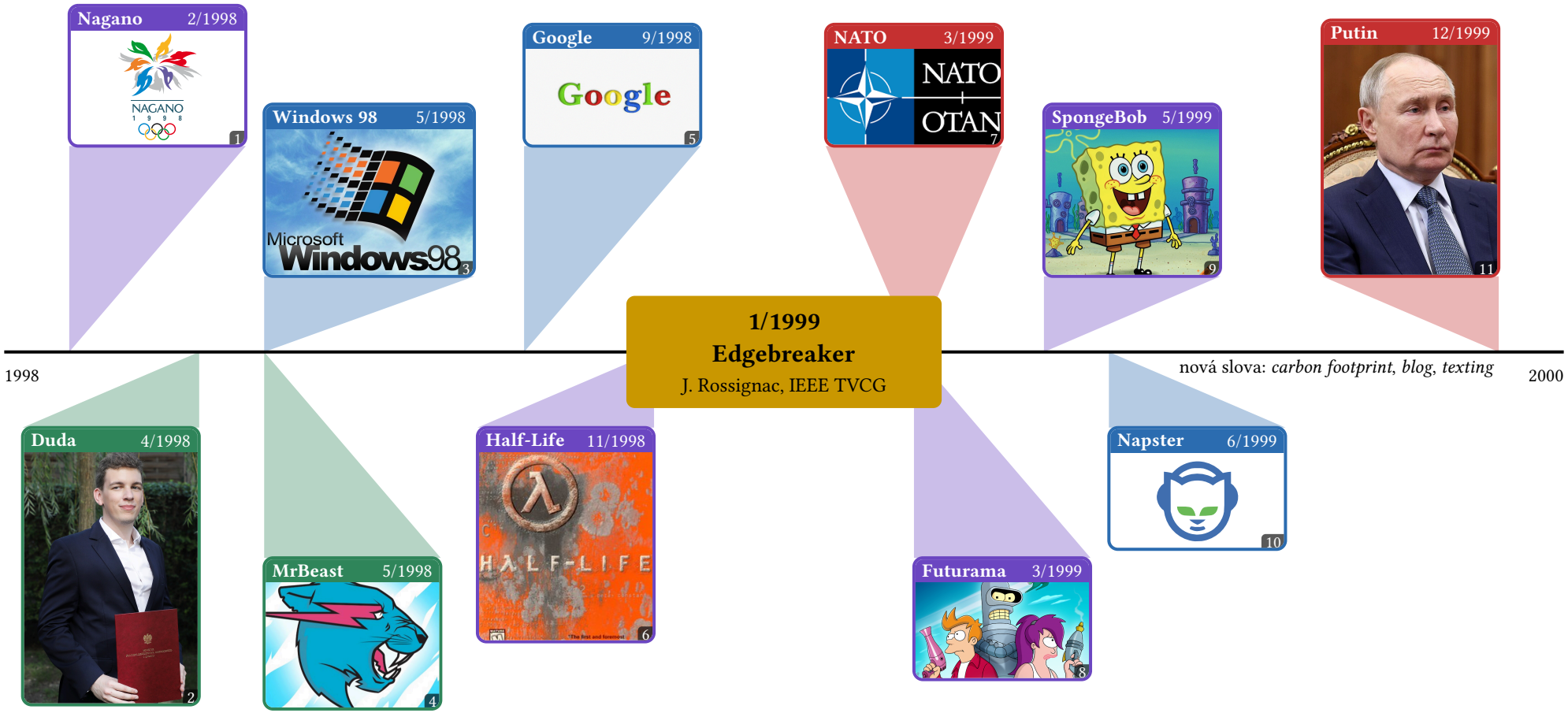


Figure 1: Dobový kontext vzniku algoritmu Edgebreaker.

1.1 Dobový kontext



Figure 2: Jarek Rossignac.

zdroj: http://kelvinkuo.com/wp-content/uploads/2010/03/KKK_6258.jpg

2. Definice problému

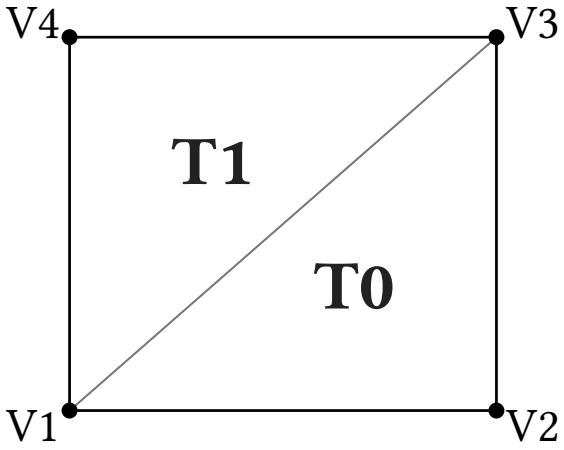


Figure 3: Trojúhelníková síť (mesh).

2.1 Mesh: geometrie + konektivita

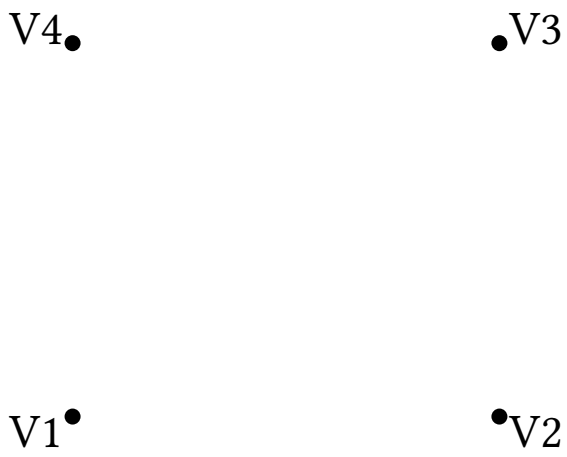


Figure 3: Trojúhelníková síť (mesh).

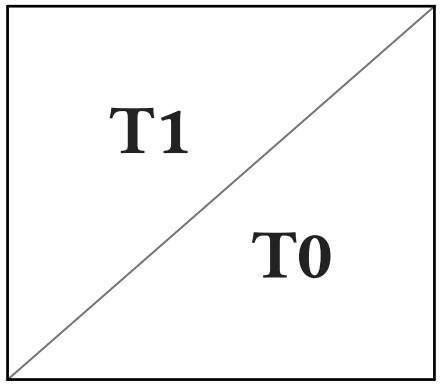


Figure 3: Trojúhelníková síť (mesh).

2.2 Jak schovat informaci?

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pořadí vertexů

[V1 V2 V3]

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pořadí vertexů

[V1 V2 V3]
[V1 V3 V2]
[V2 V1 V3]
[V3 V1 V2]
[V2 V3 V1]
[V3 V2 V1]

2.2 Jak schovat informaci?

pořadí vertexů

[V1 V2 V3]
[V1 V3 V2]
[V2 V1 V3]
[V3 V1 V2]
[V2 V3 V1]
[V3 V2 V1]

CLERS řetězec

2.2 Jak schovat informaci?

pořadí vertexů	CLERS řetězec
[V1 V2 V3]	CLRSLRCLEE
[V1 V3 V2]	
[V2 V1 V3]	
[V3 V1 V2]	
[V2 V3 V1]	
[V3 V2 V1]	

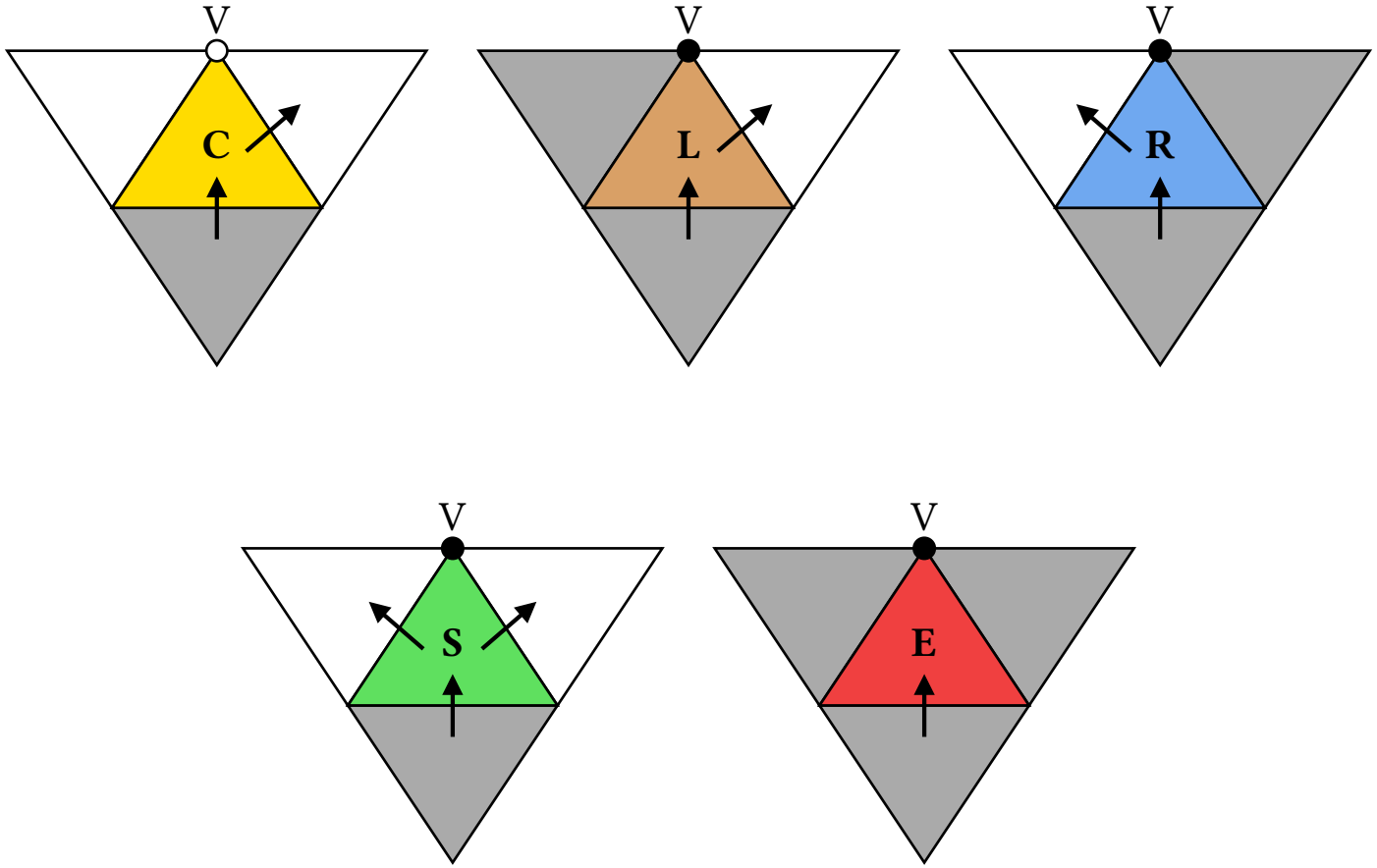


Figure 4: Základní CLERS operace.

3. Komprese

3.1 Omezení

Mesh musí být **orientovatelný manifold** homeomorfní s koulí/kruhem.

Lehce upravená verze algoritmu zvládne i *díry* a *handly*, obecně meshe s vyšším genusem.

History =

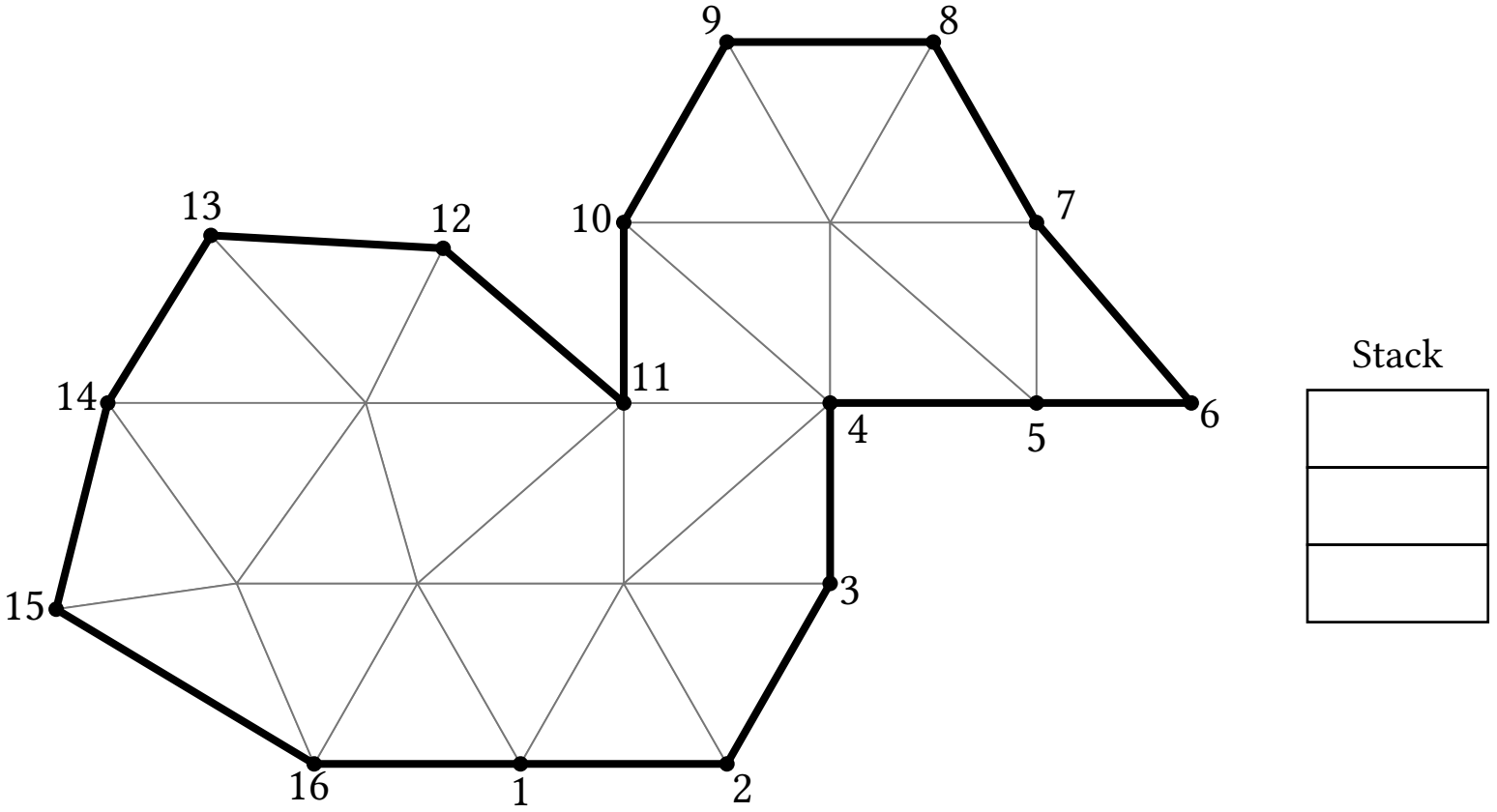


Figure 5: Jednoduchá komprese.

History = C

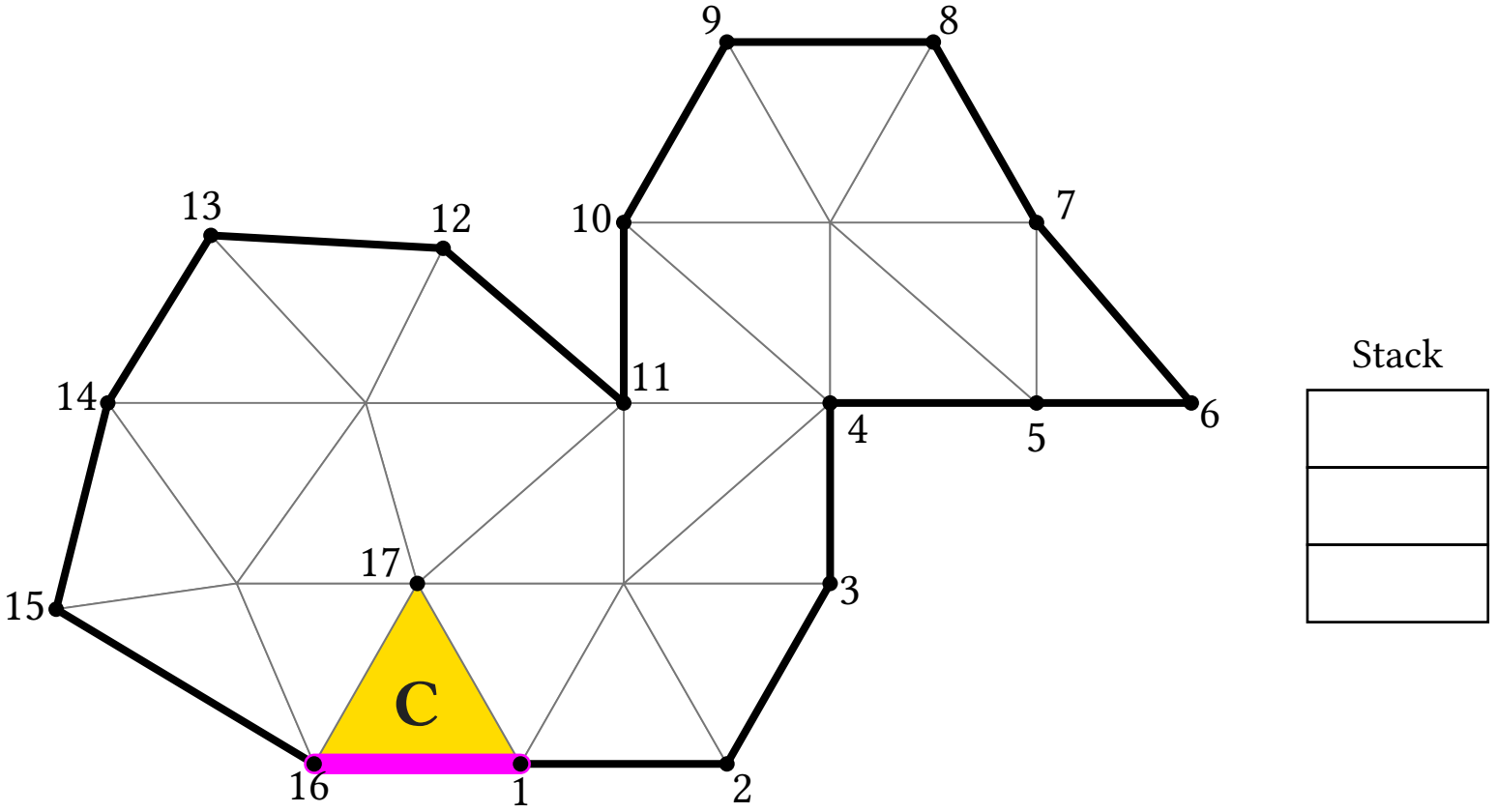


Figure 5: Jednoduchá komprese.

History = CC

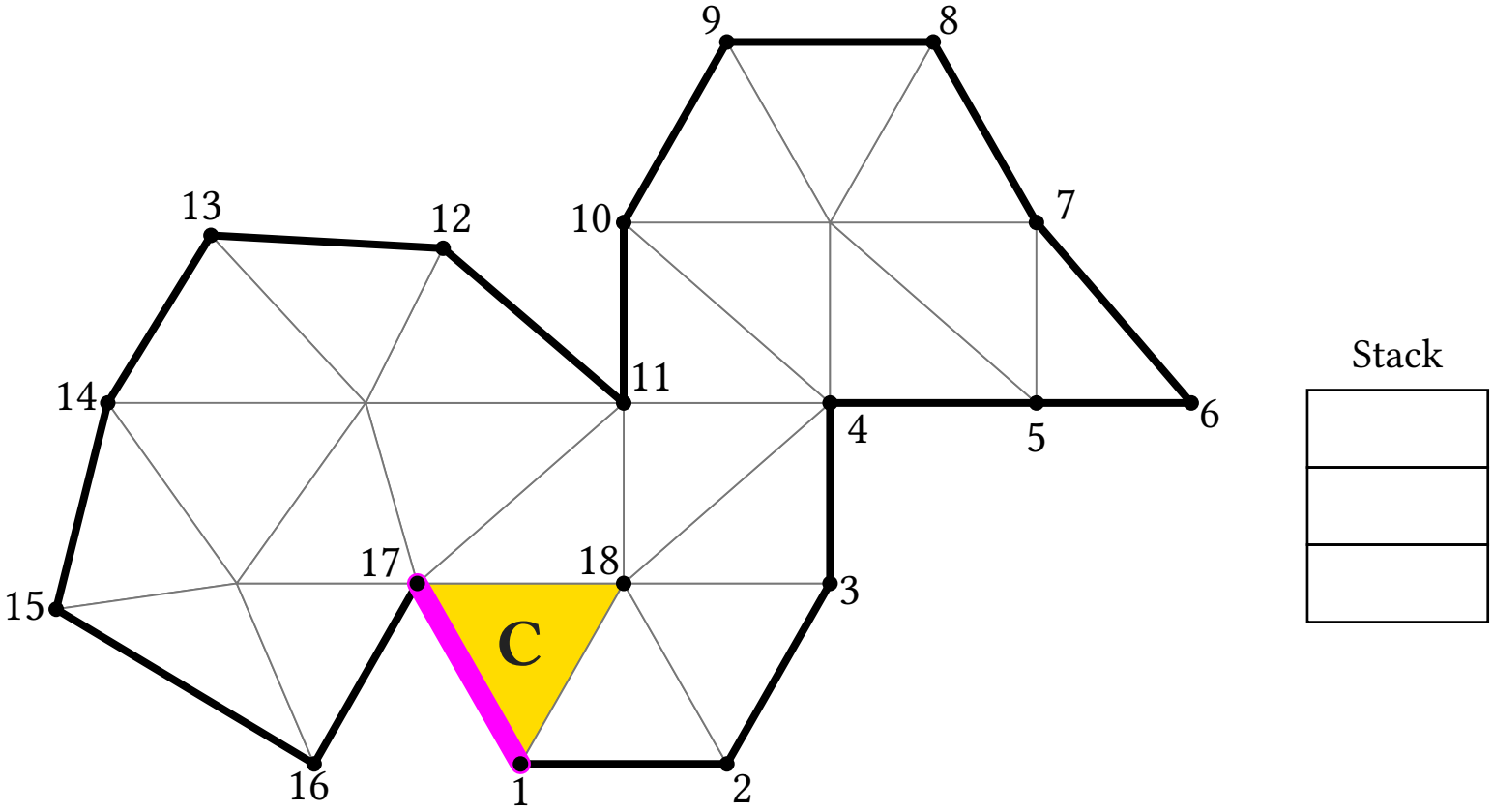


Figure 5: Jednoduchá komprese.

History = CCR

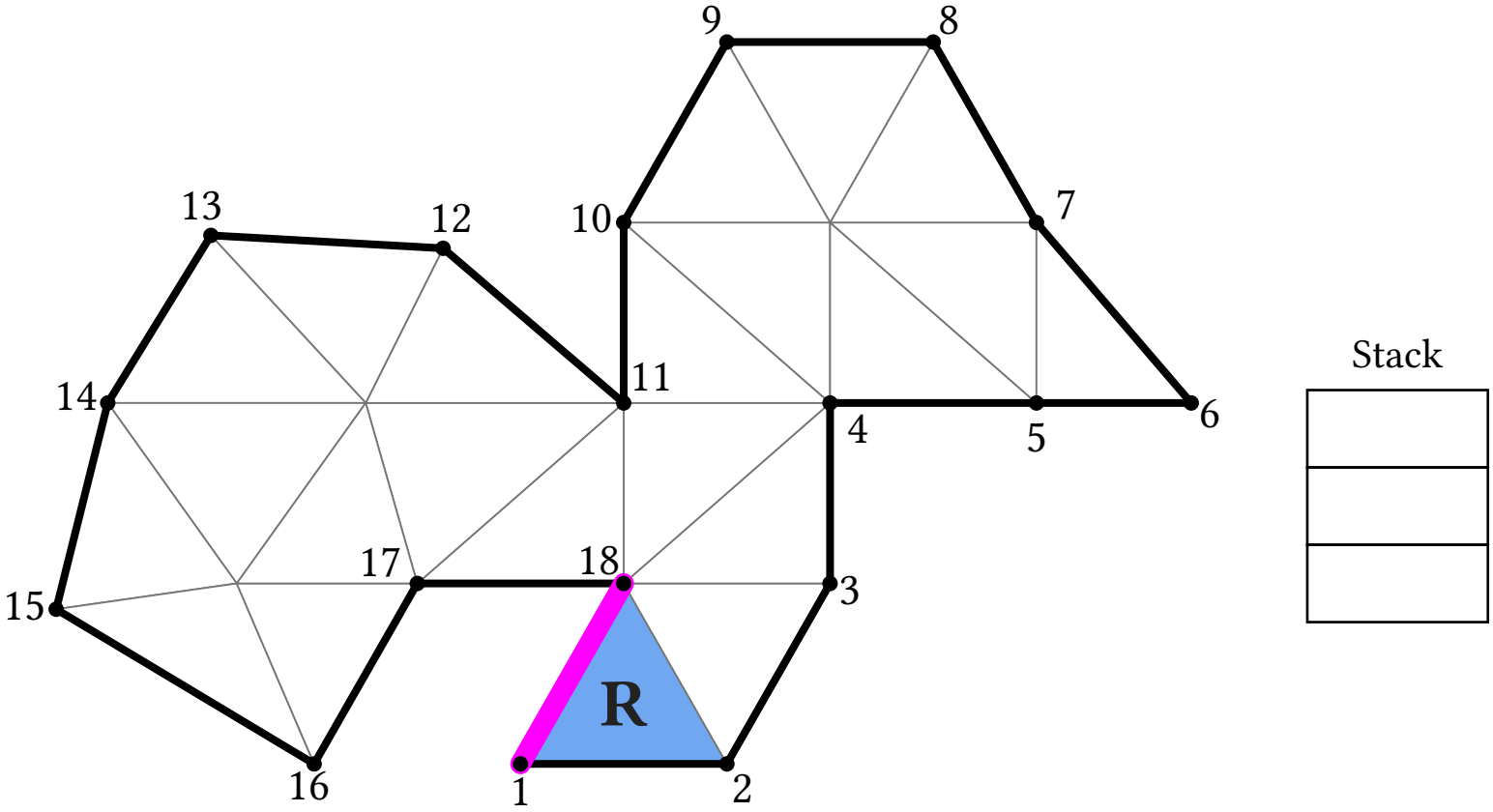


Figure 5: Jednoduchá komprese.

History = CCRR

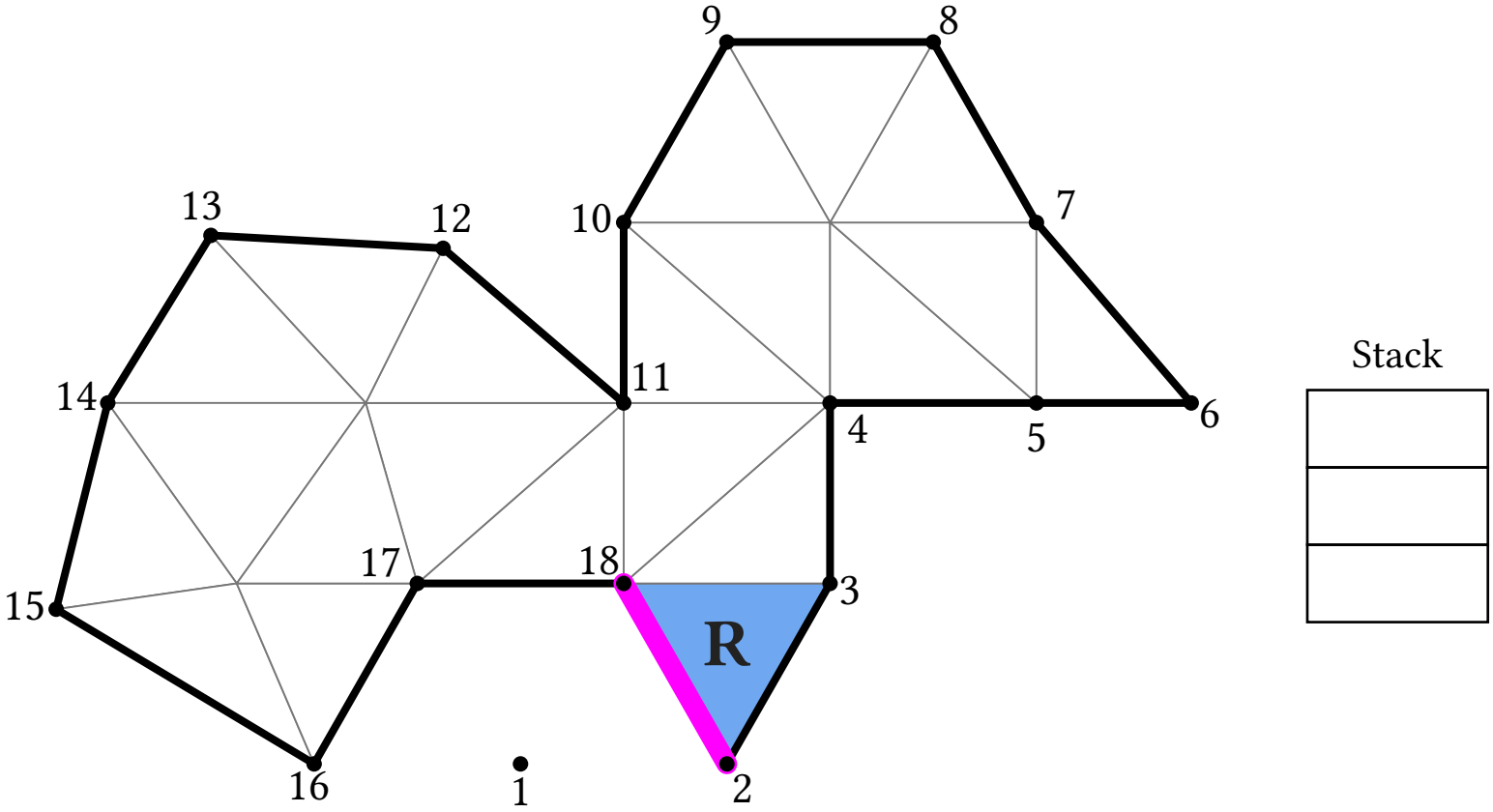


Figure 5: Jednoduchá komprese.

History = CCRRR

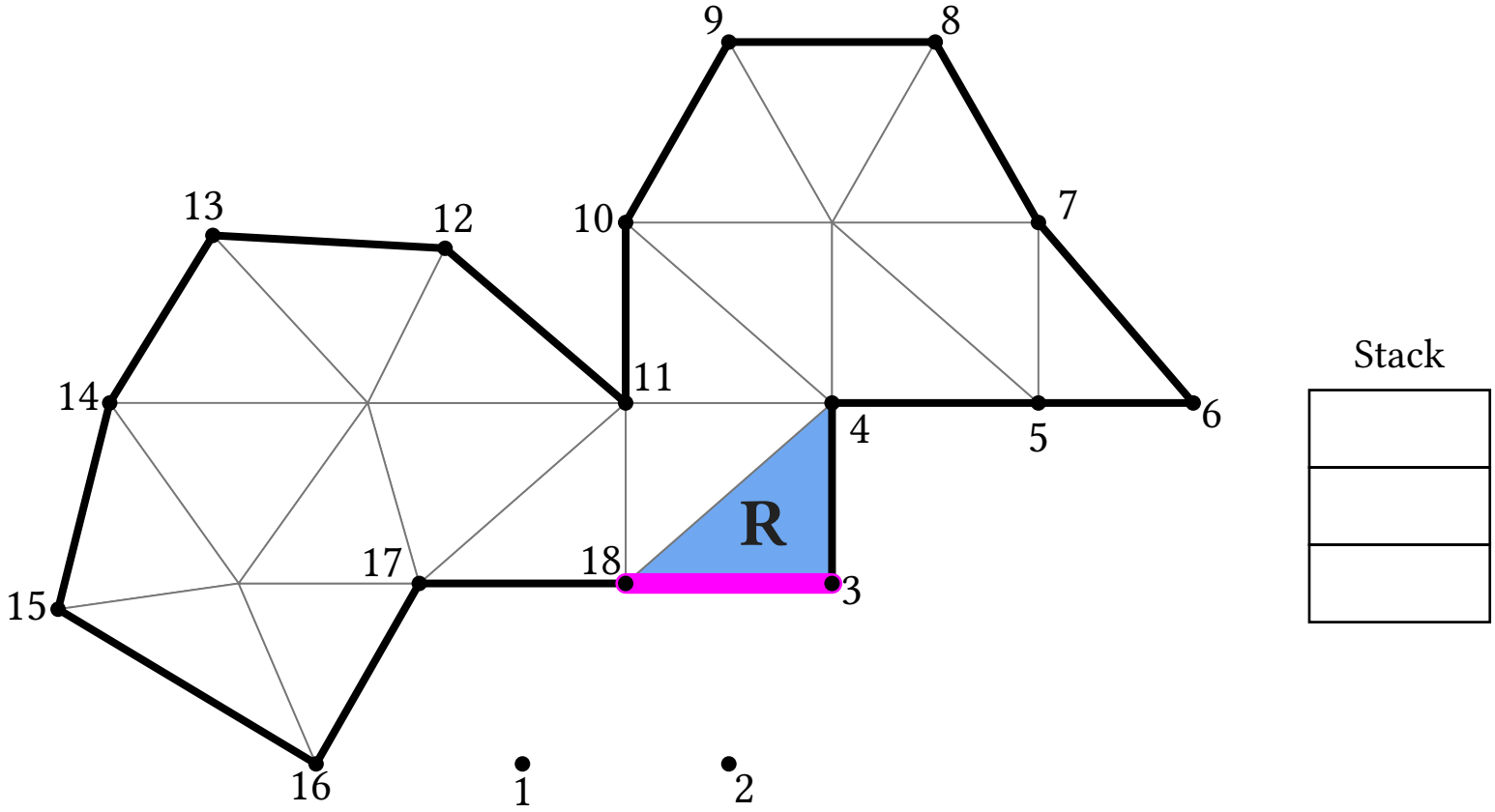


Figure 5: Jednoduchá komprese.

History = CCRRRS

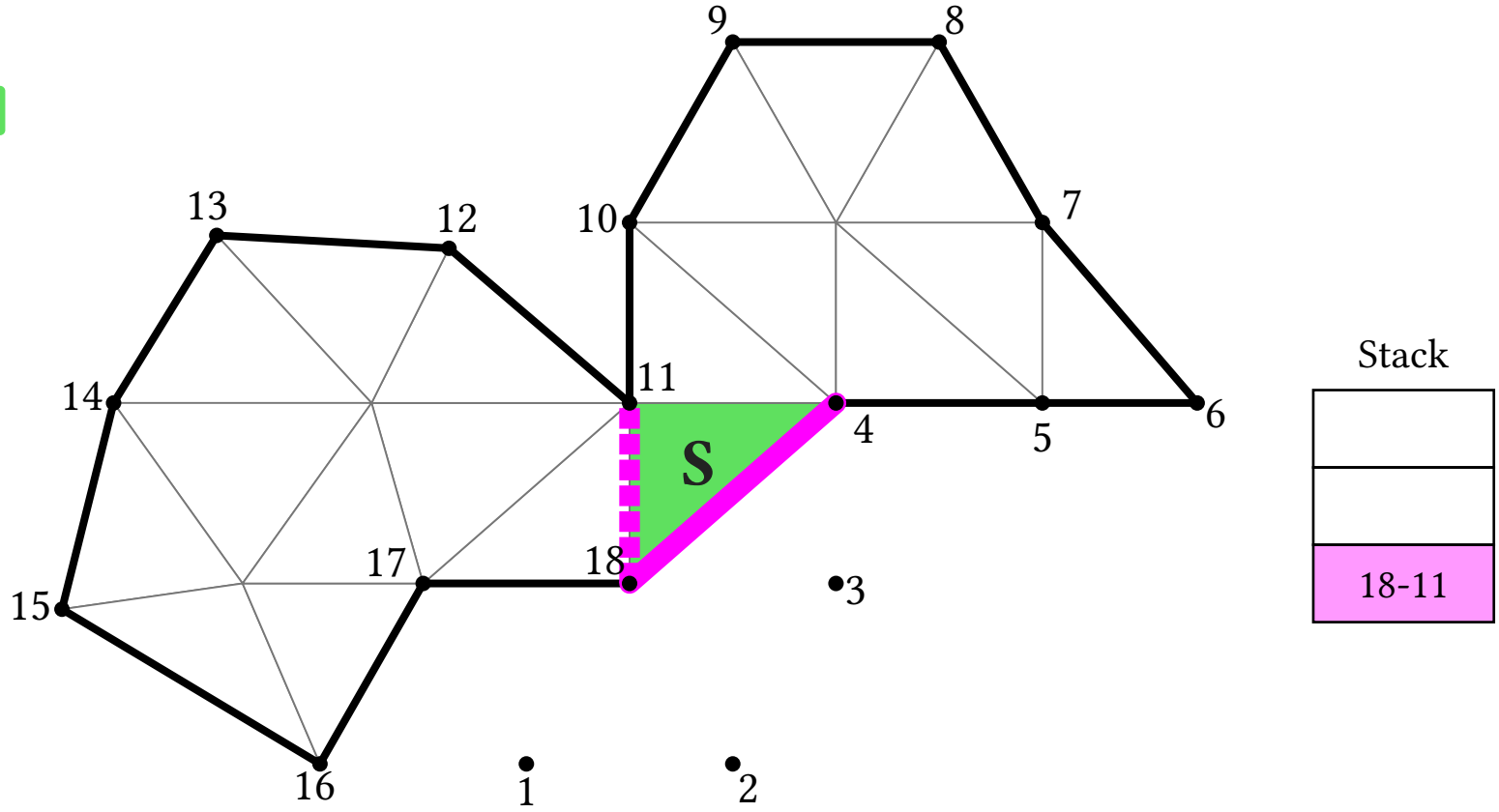


Figure 5: Jednoduchá komprese.

History = CCRRRSL

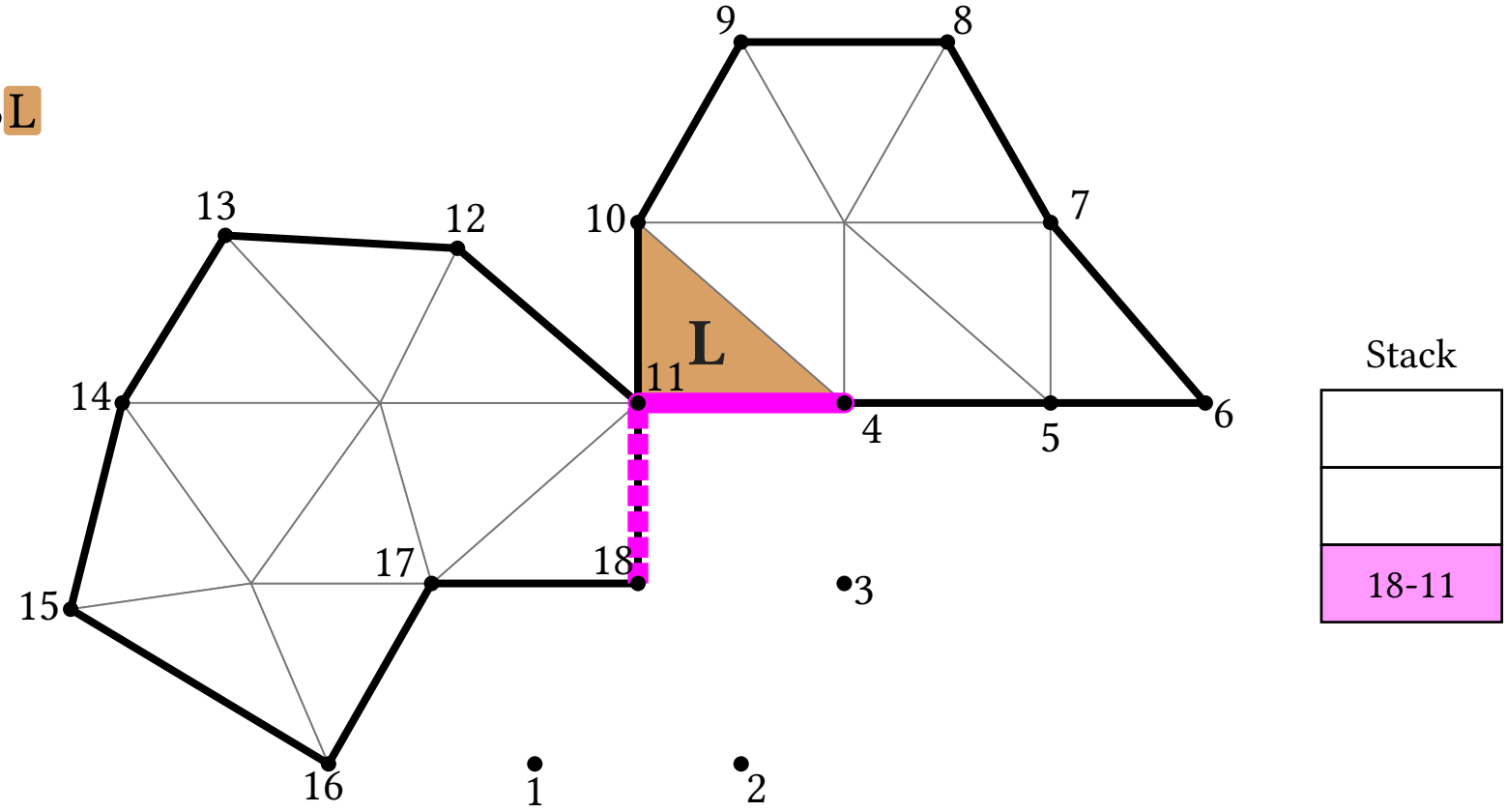


Figure 5: Jednoduchá komprese.

History = CCRRRSLC

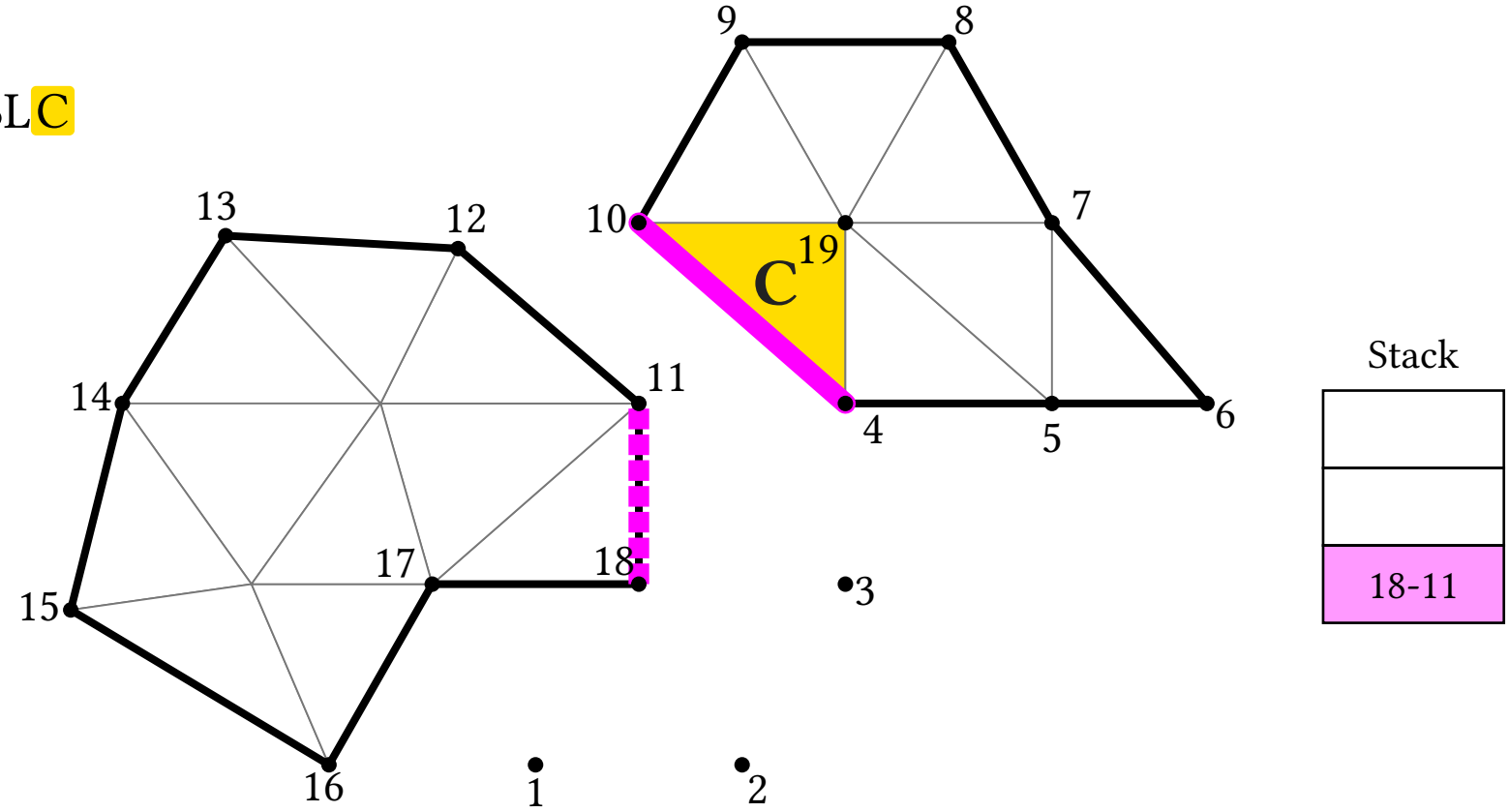


Figure 5: Jednoduchá komprese.

History = CCRRRSLCR

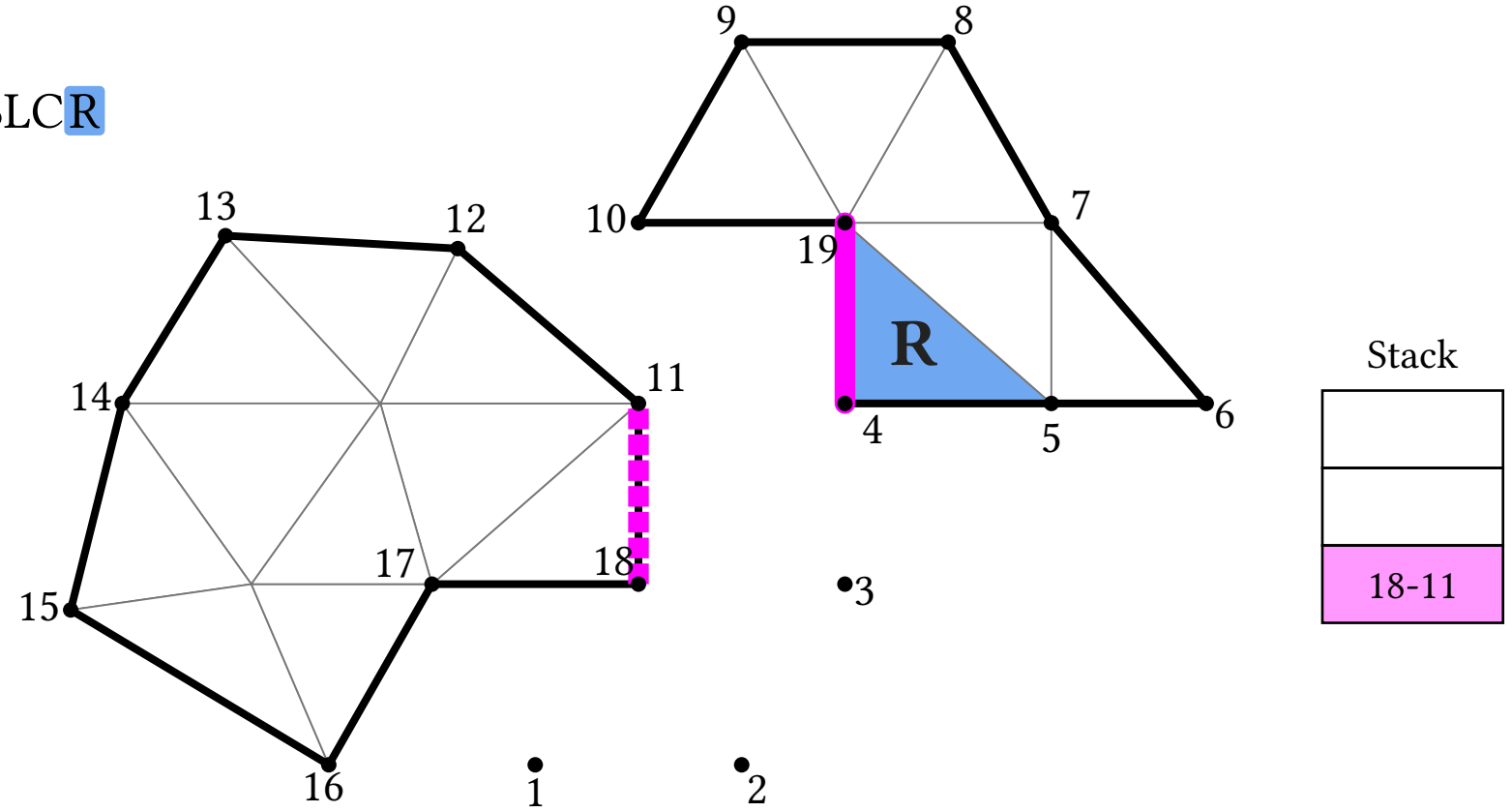


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRS

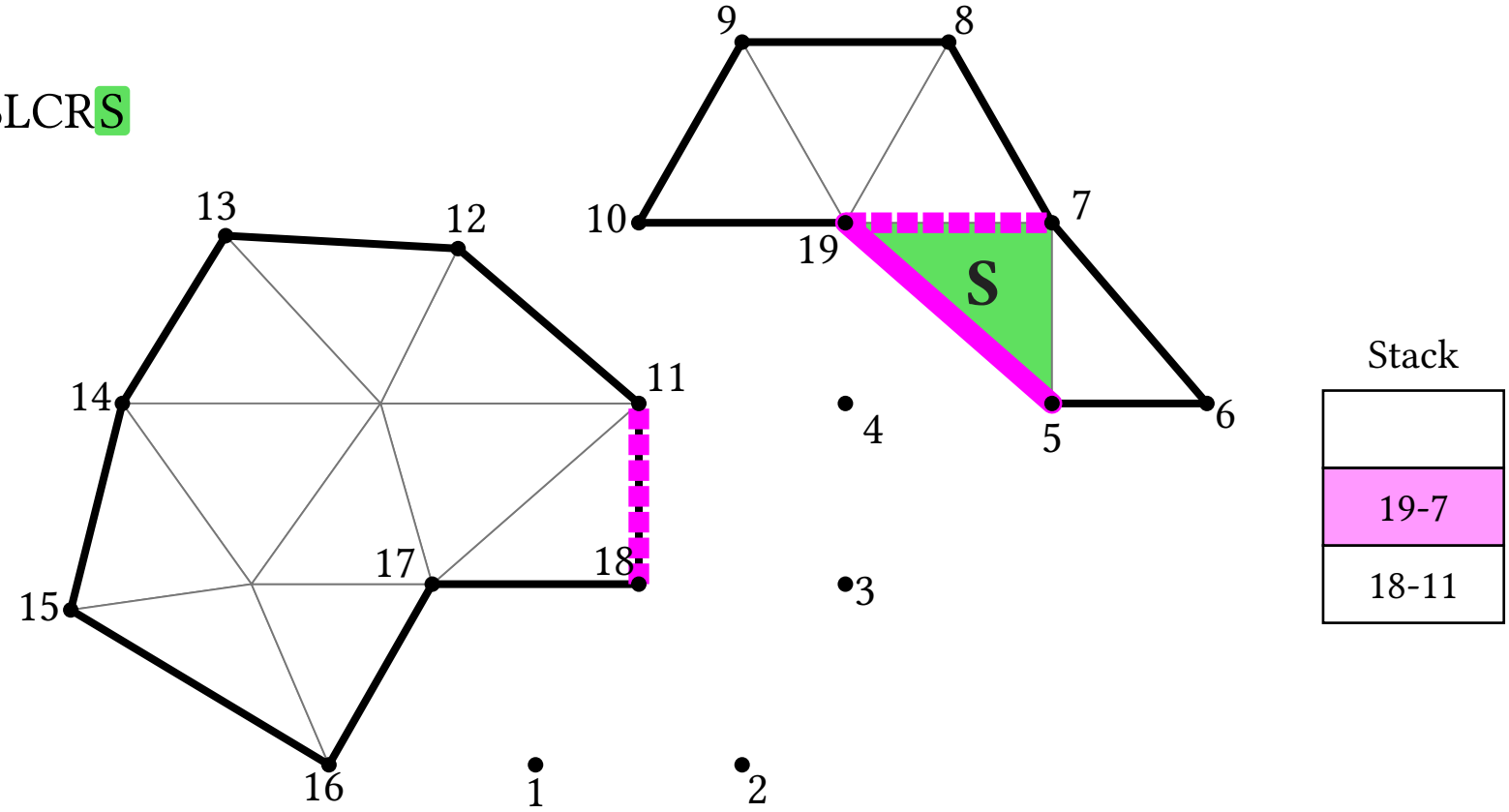


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSE

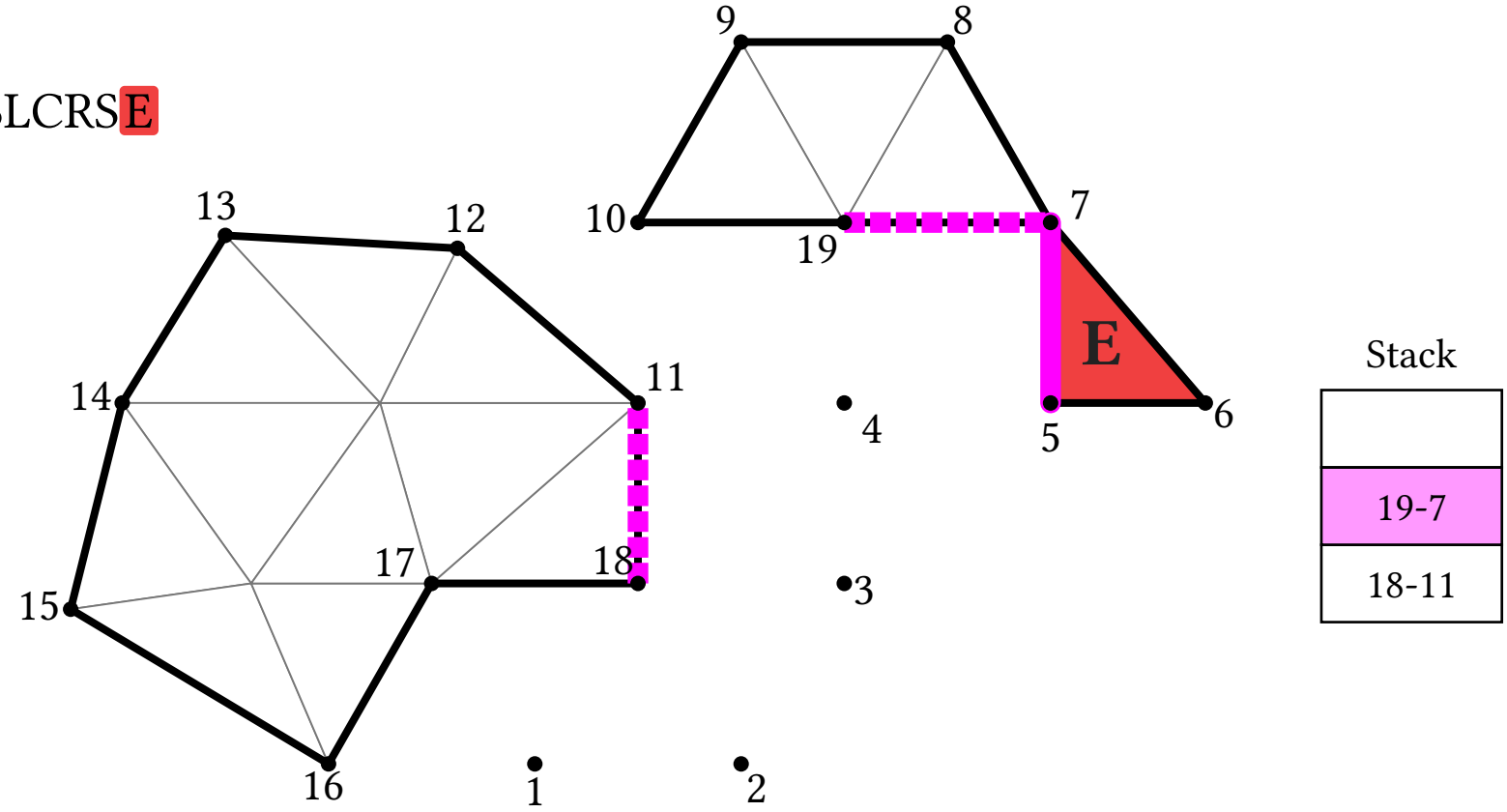


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSER

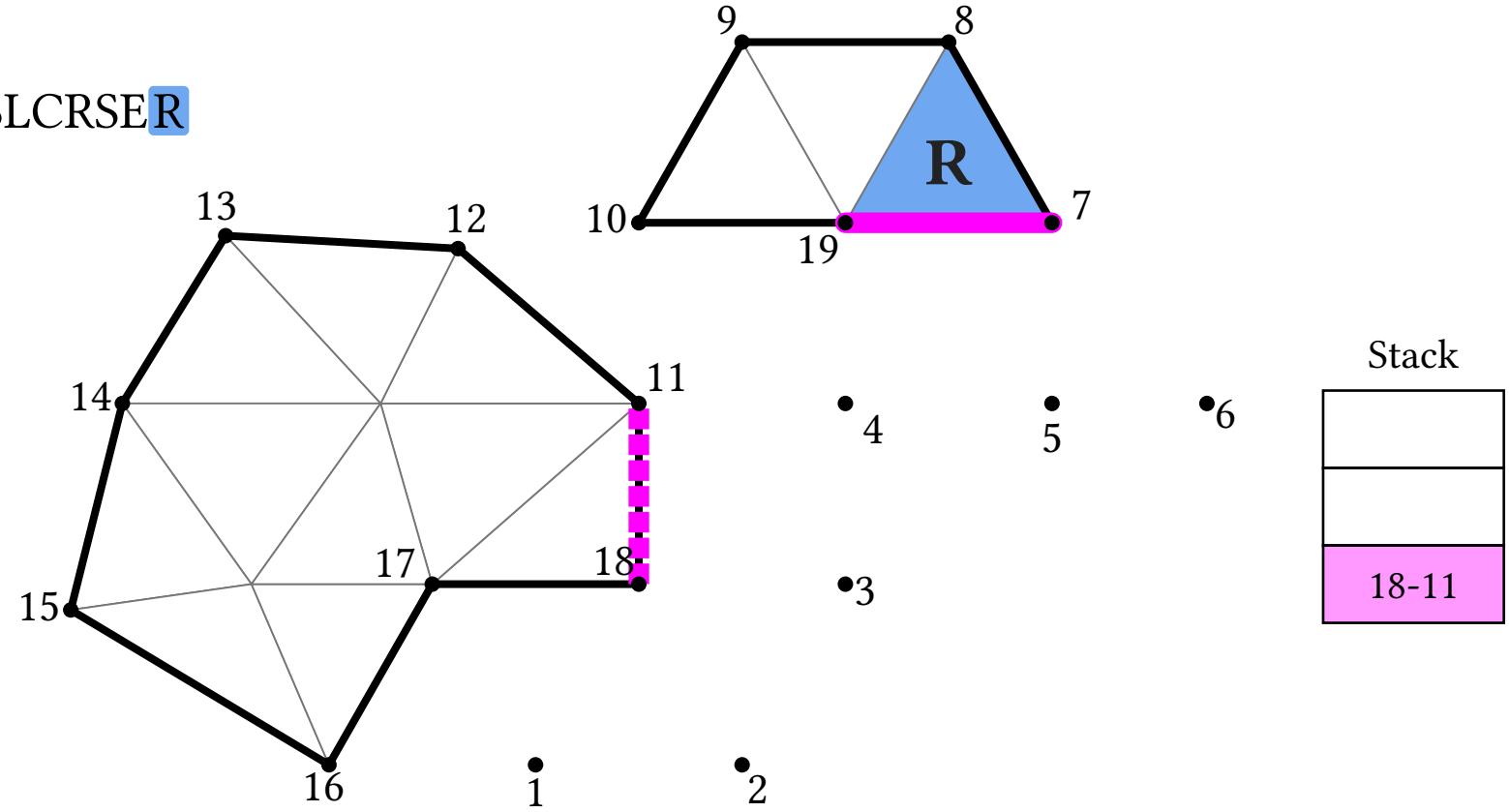


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERR

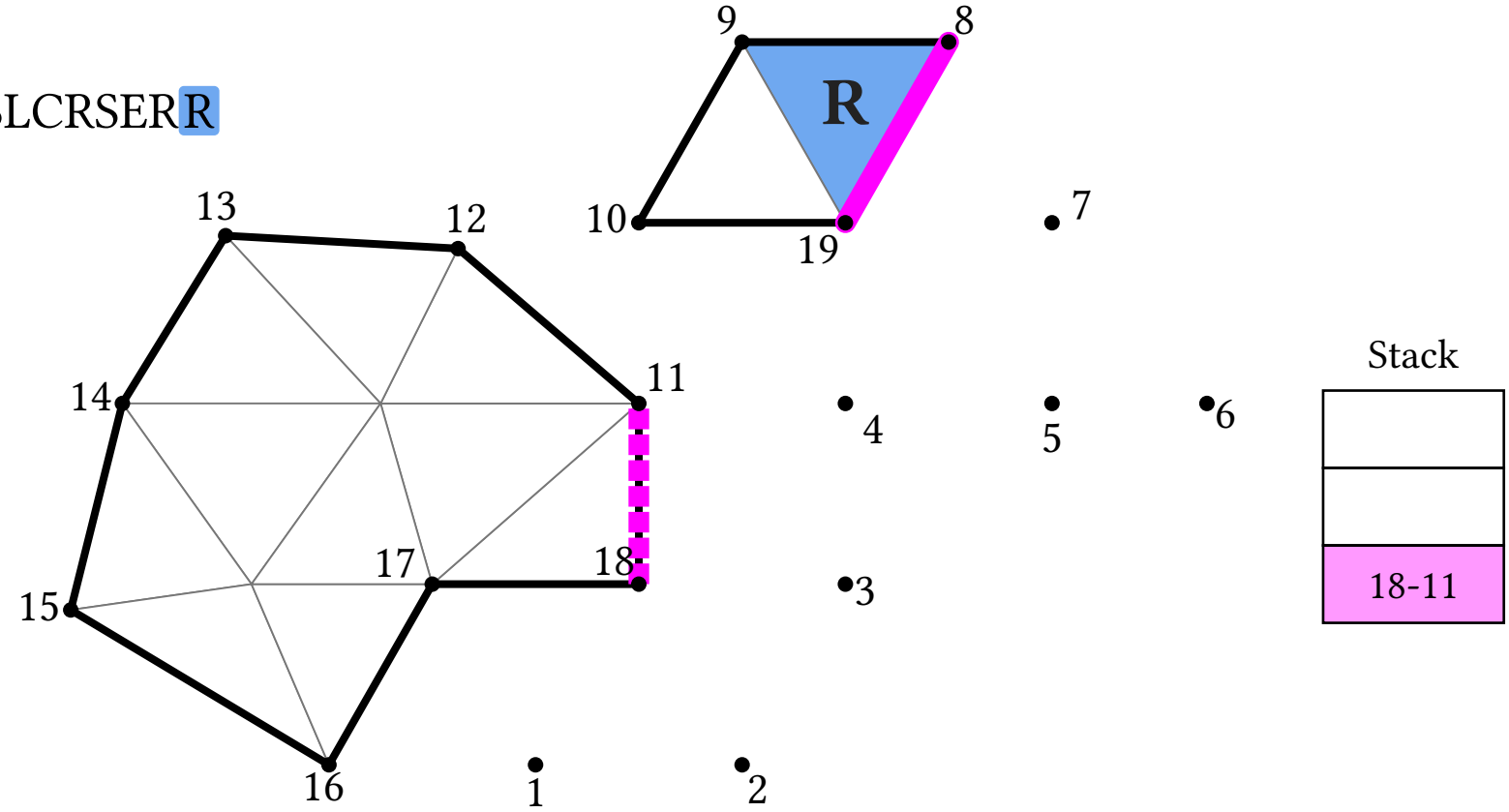


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRE

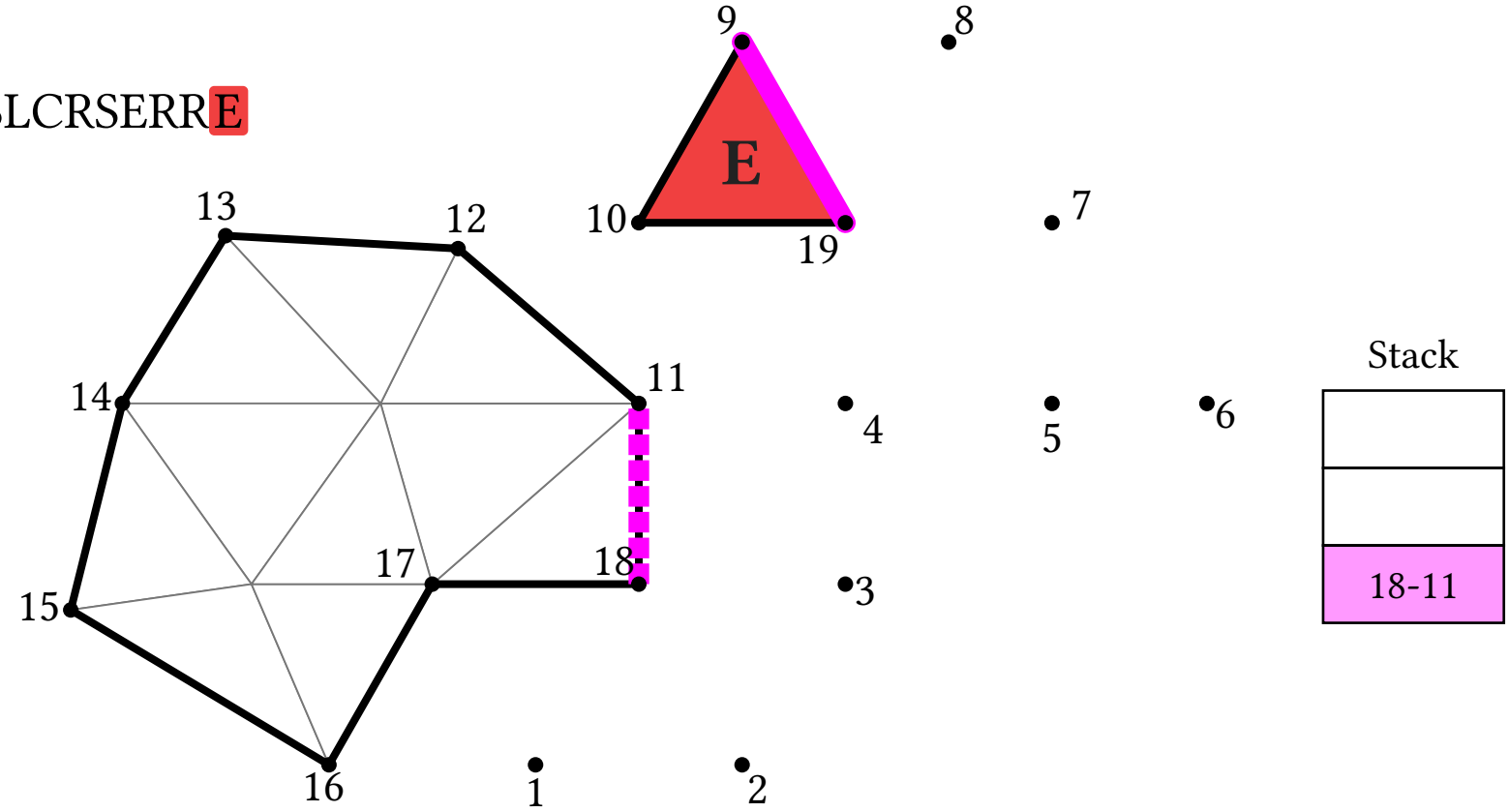


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERREL

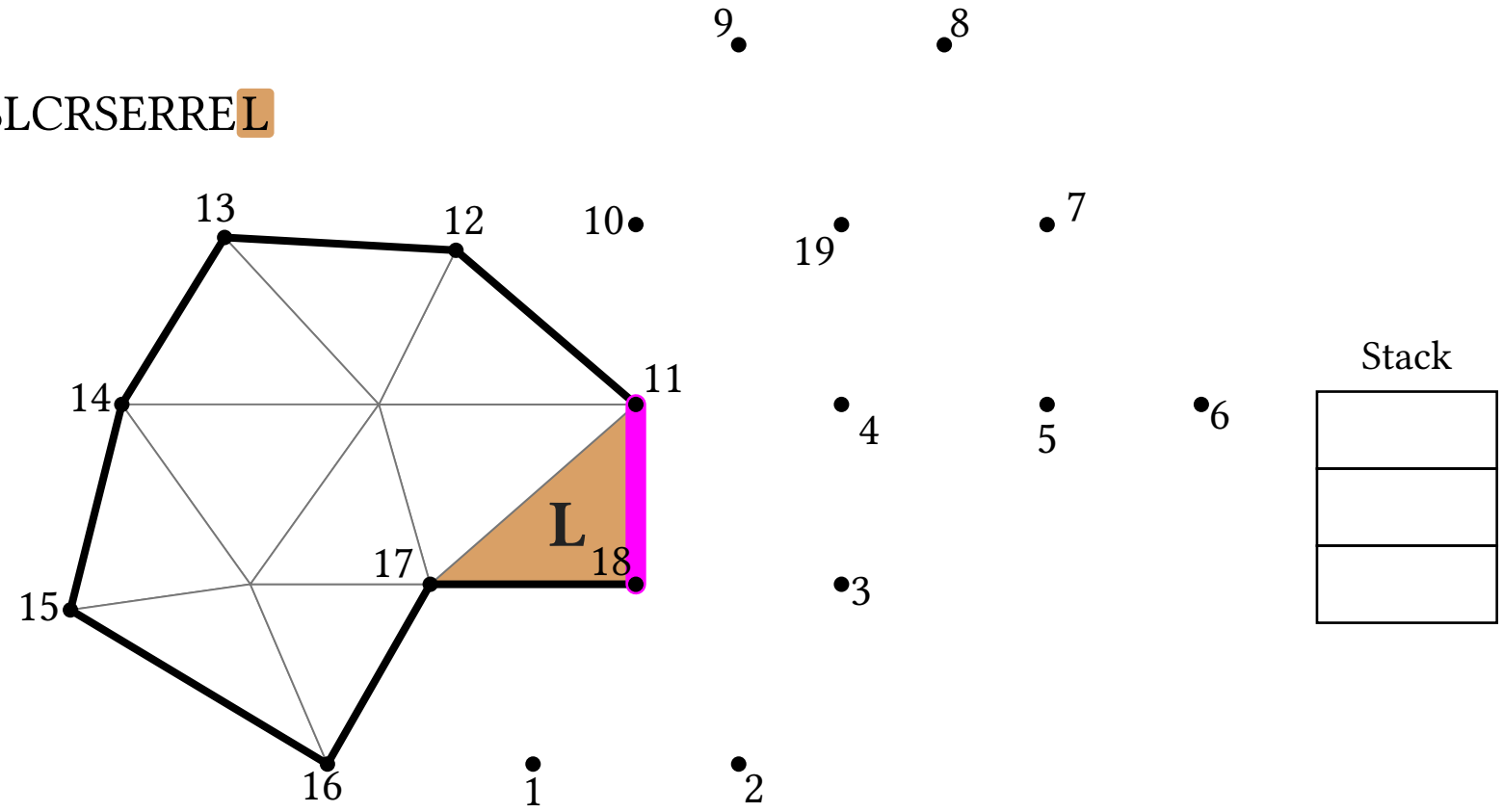


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERREL**C**

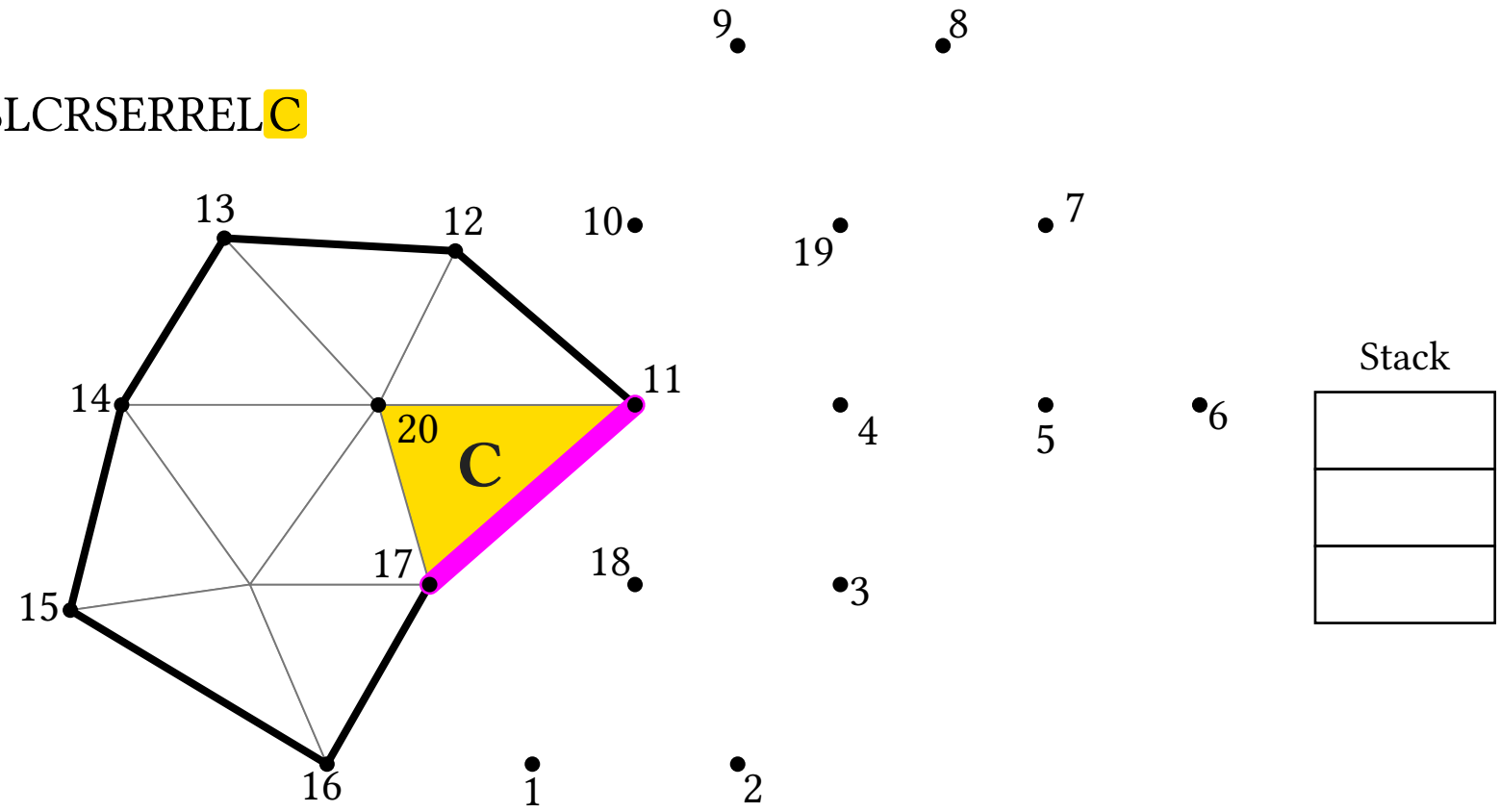


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCR

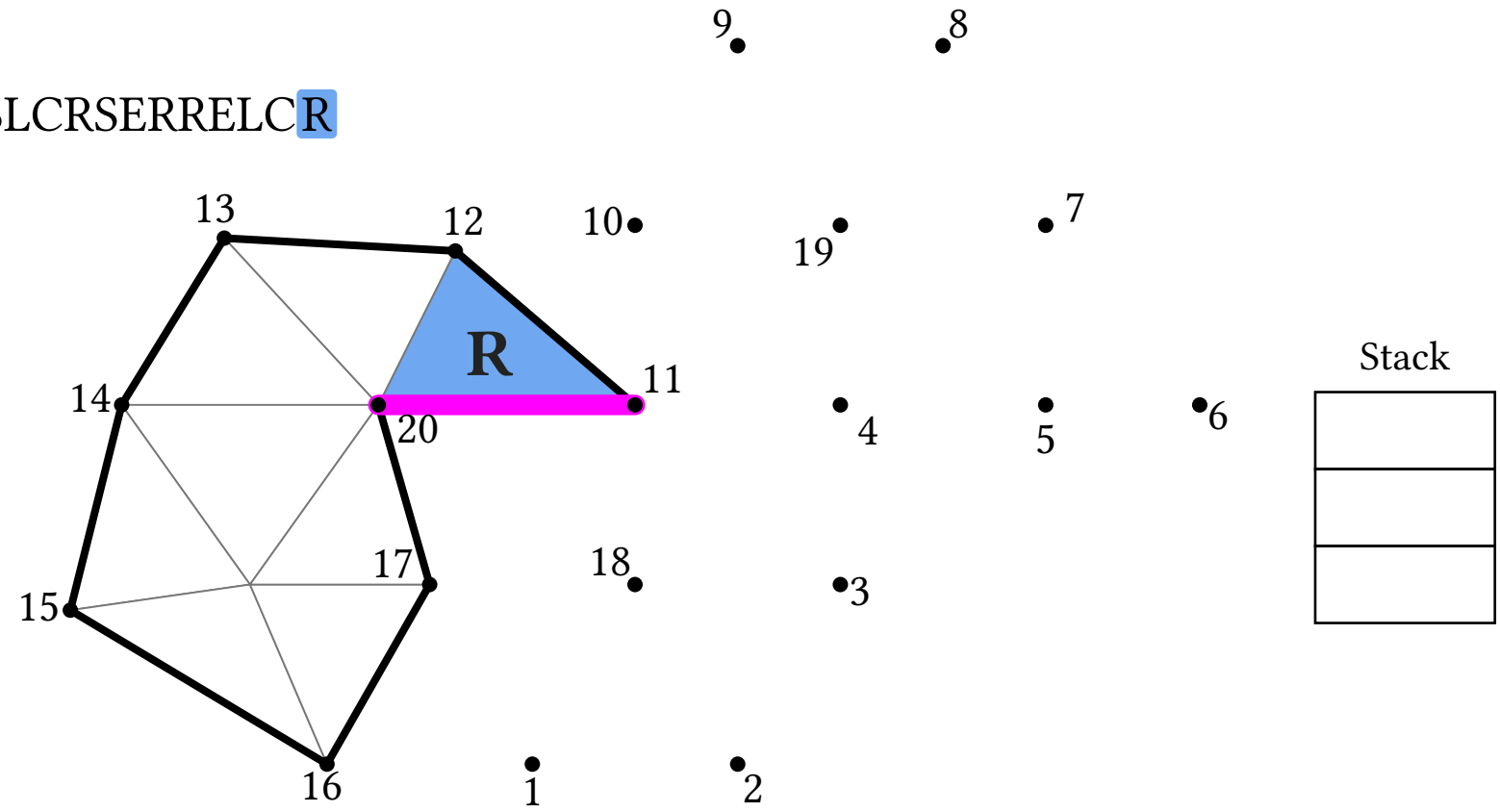


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRR

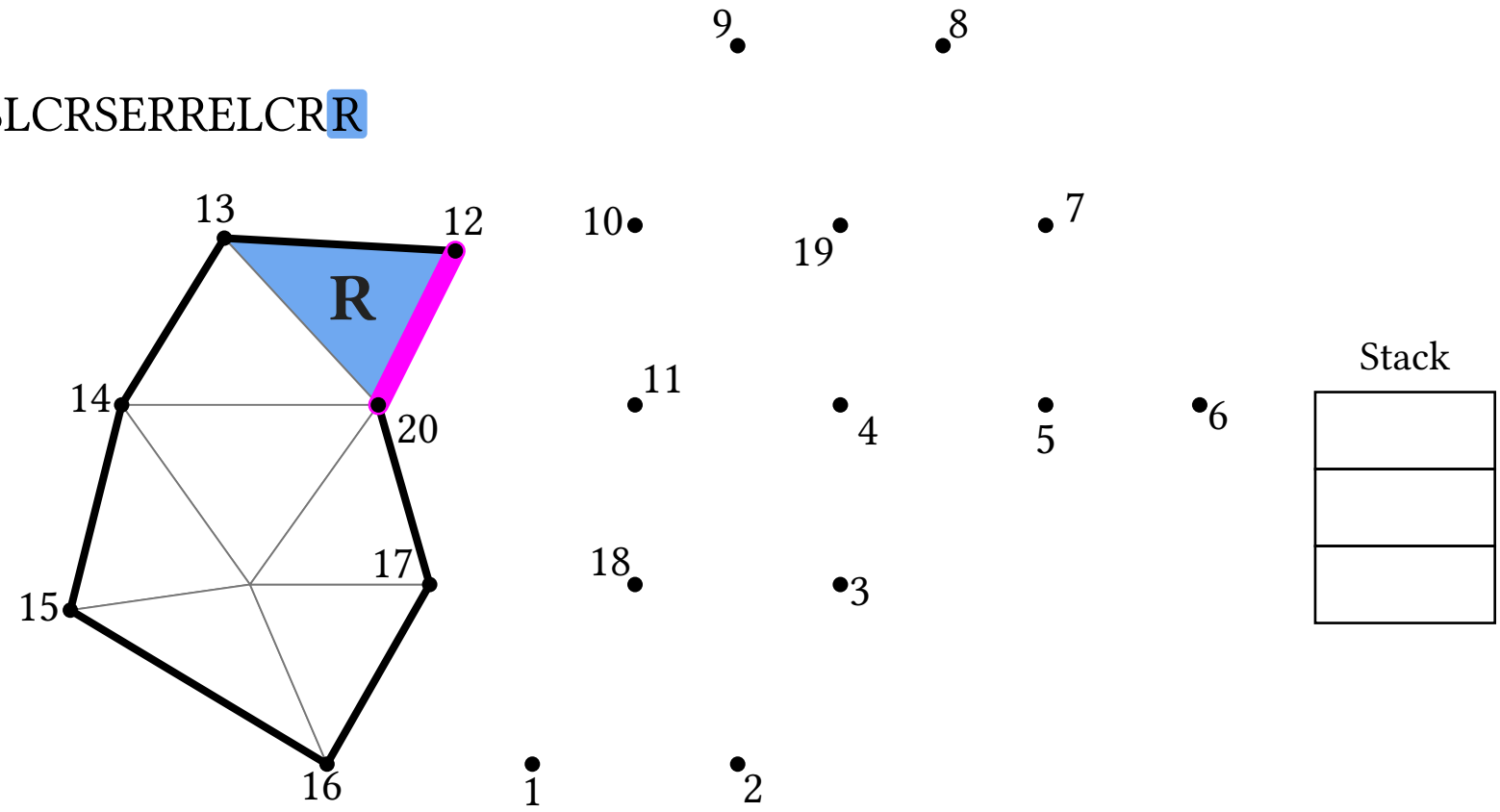


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRR

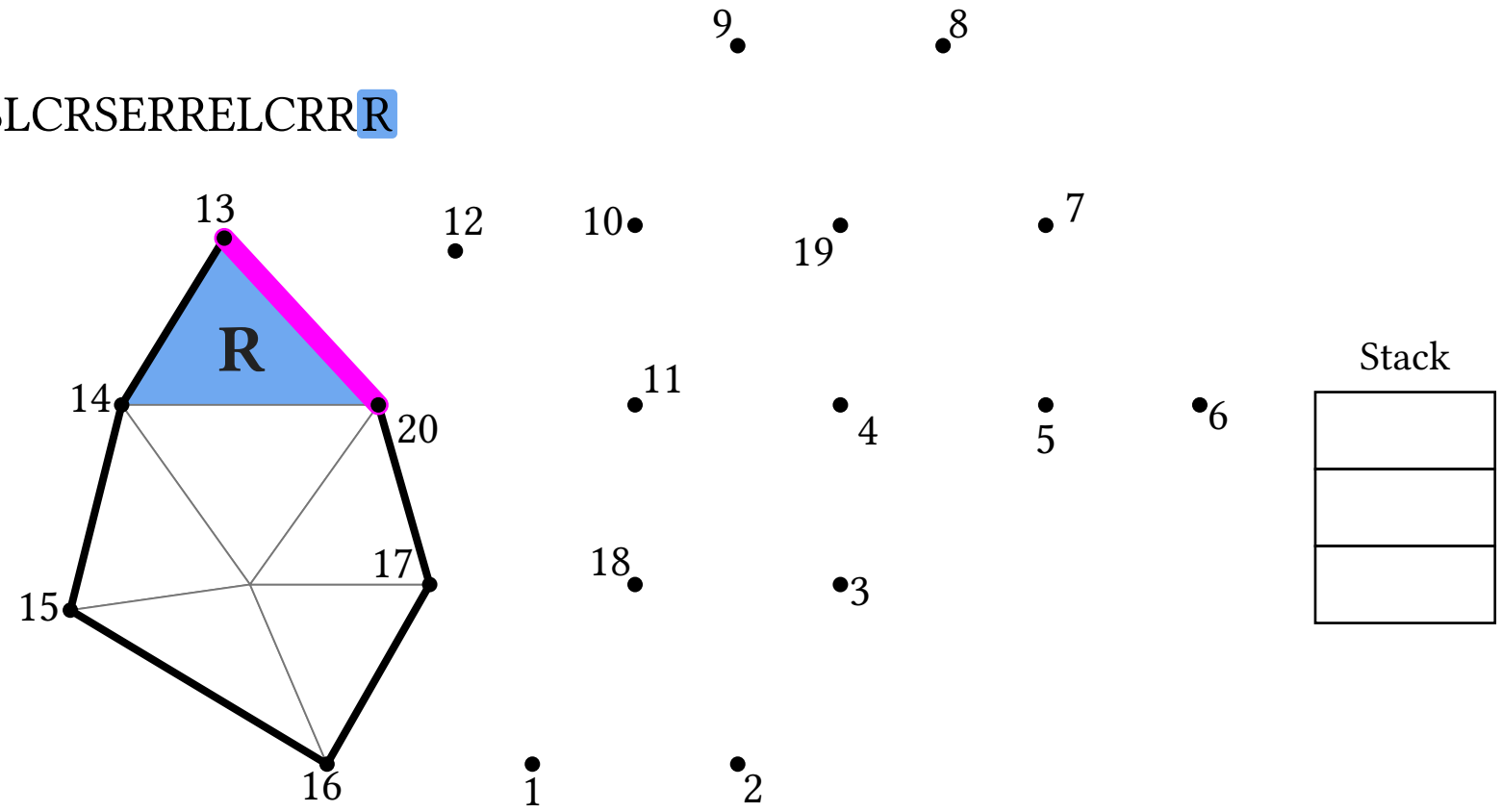


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRC

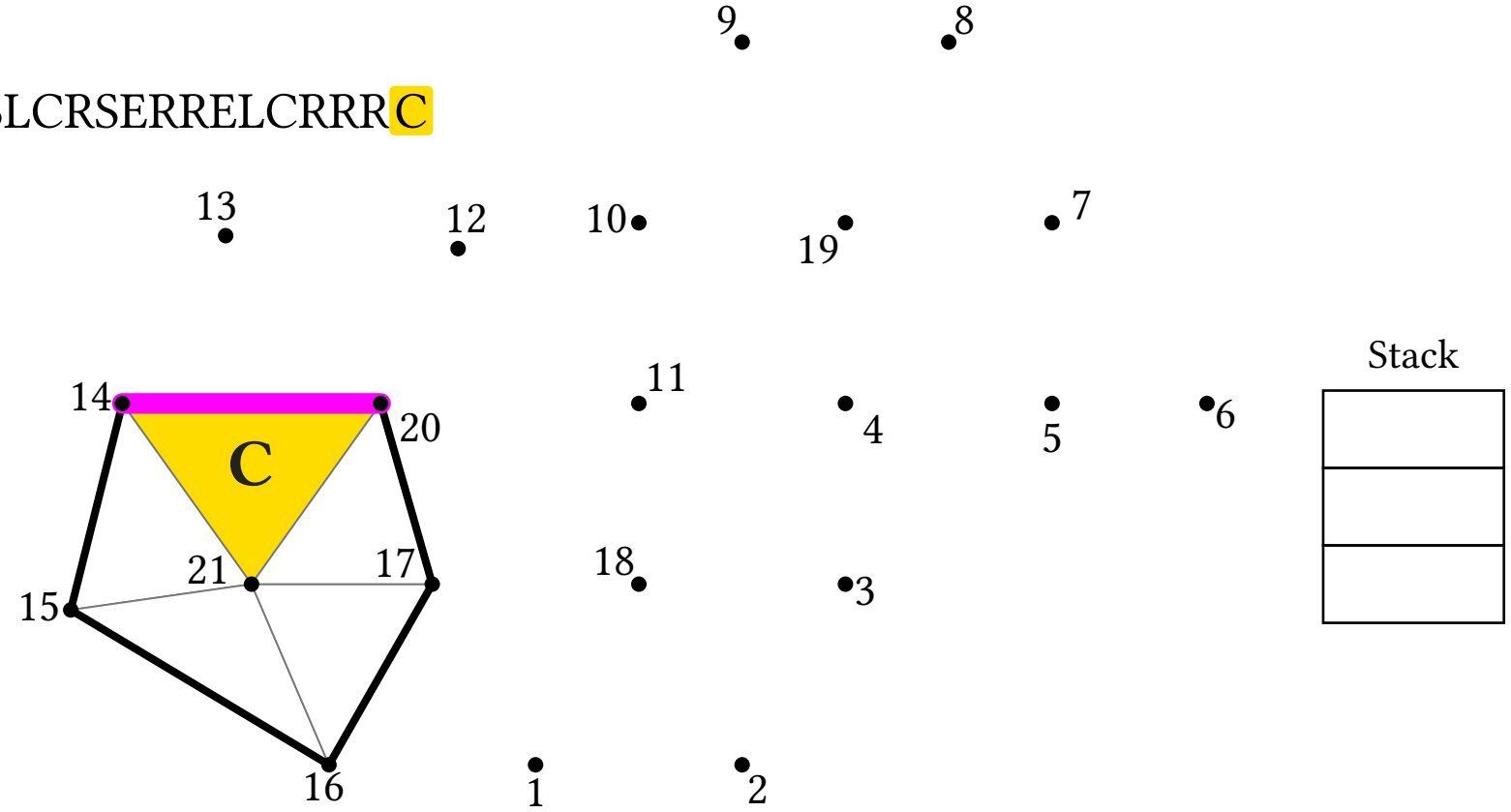


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCR

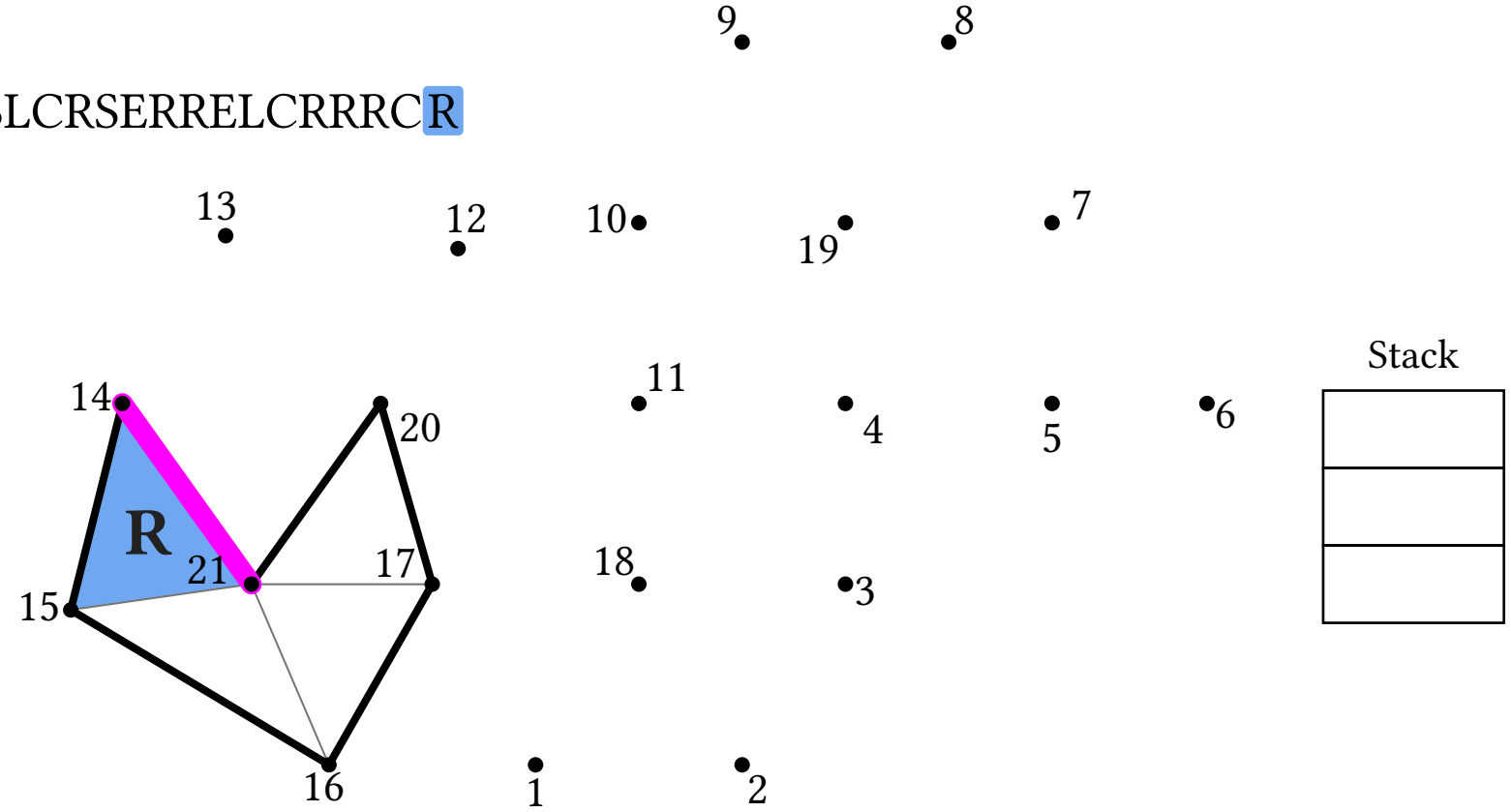


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRR

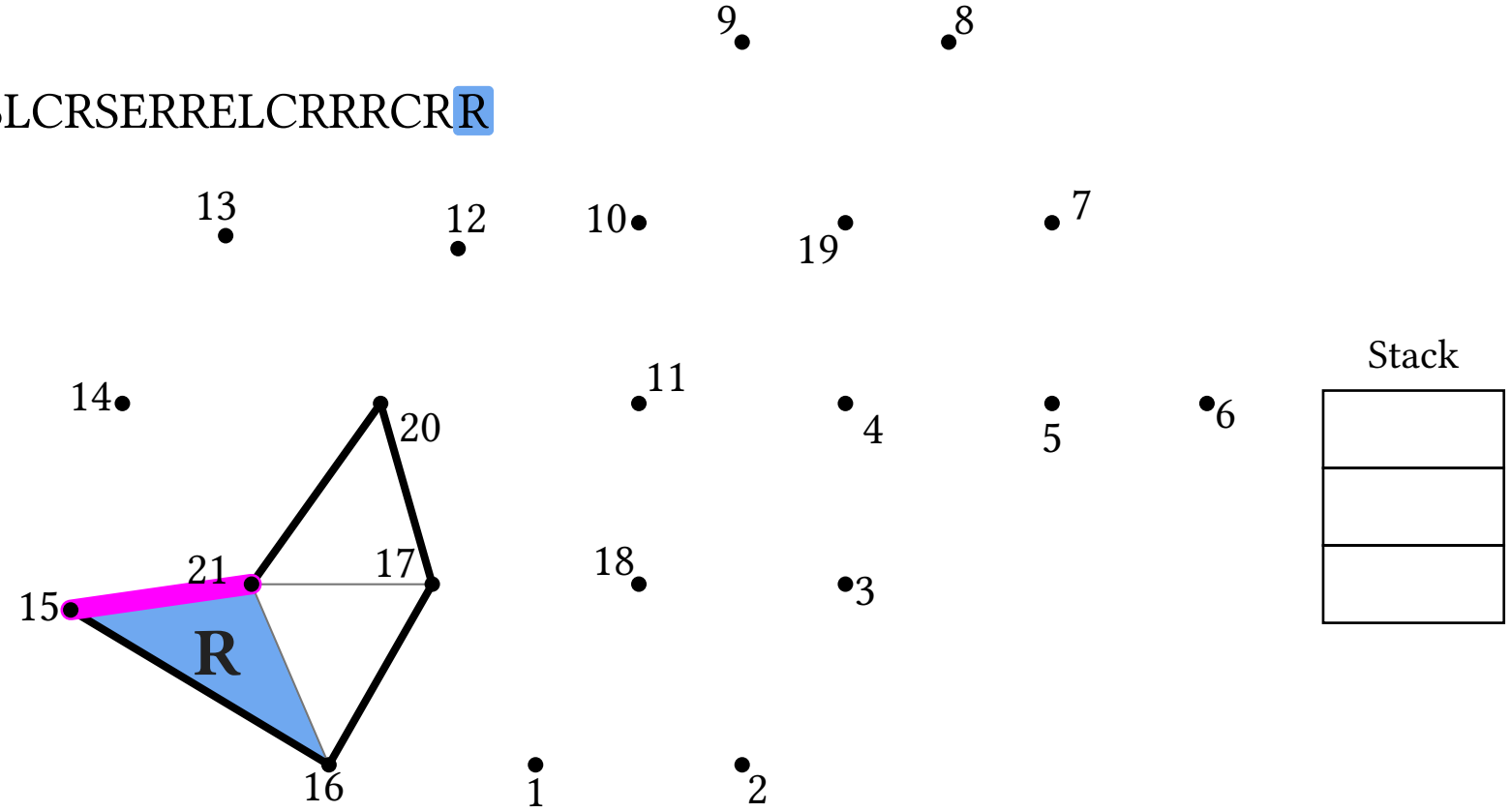


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRR

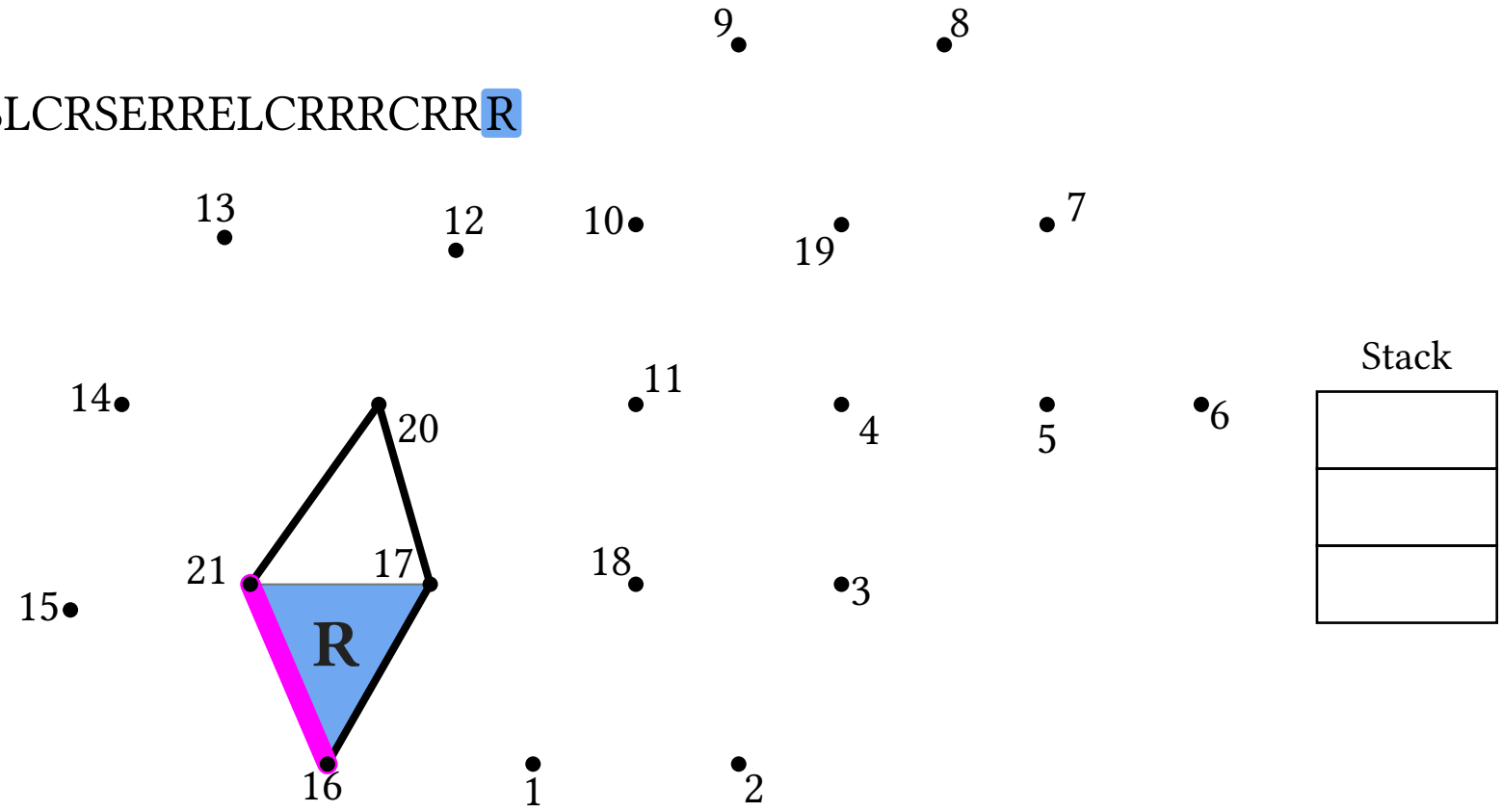


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRRE

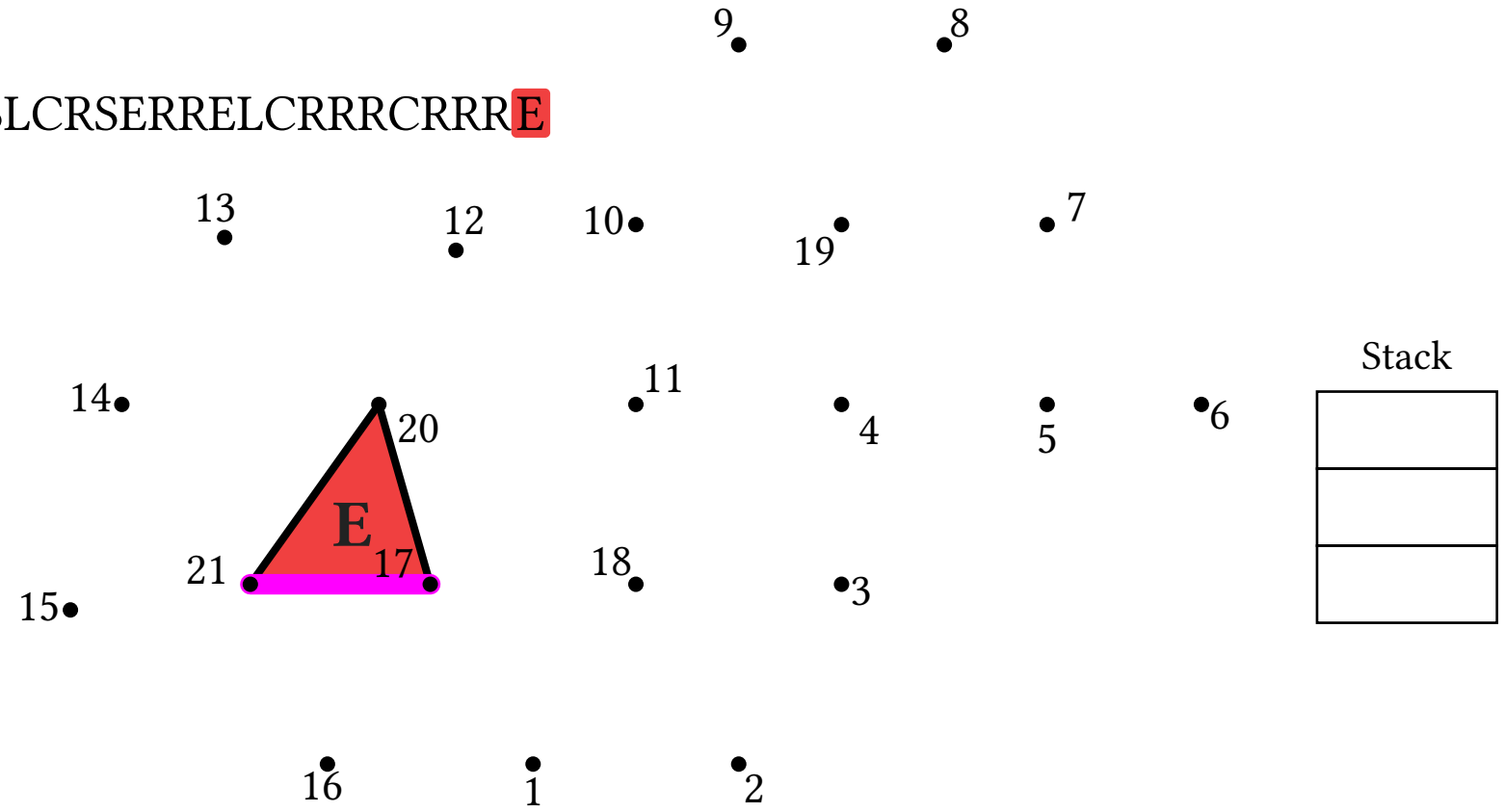


Figure 5: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRRE

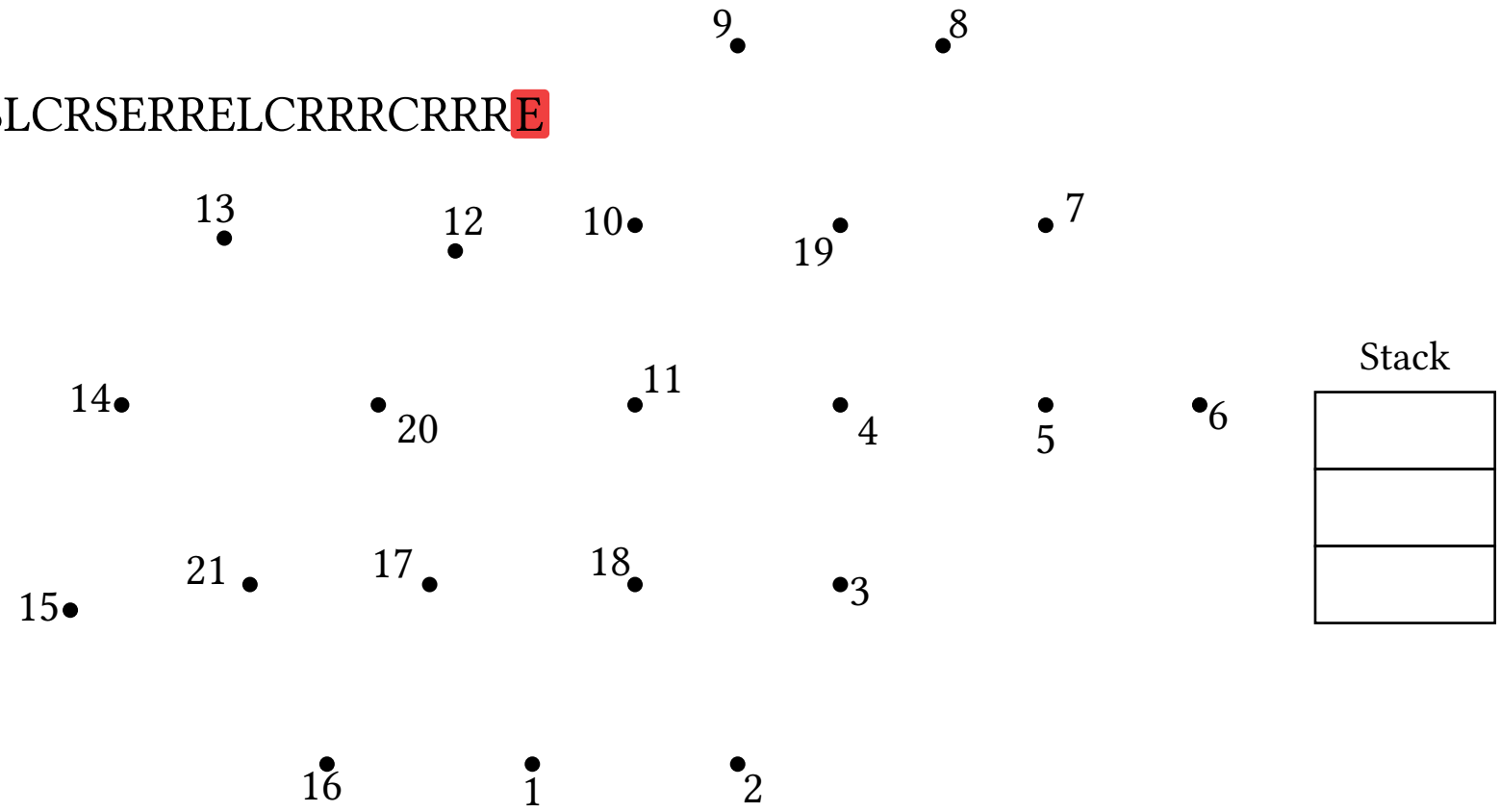


Figure 5: Jednoduchá komprese.

3.3 Vyšší genus?

History =

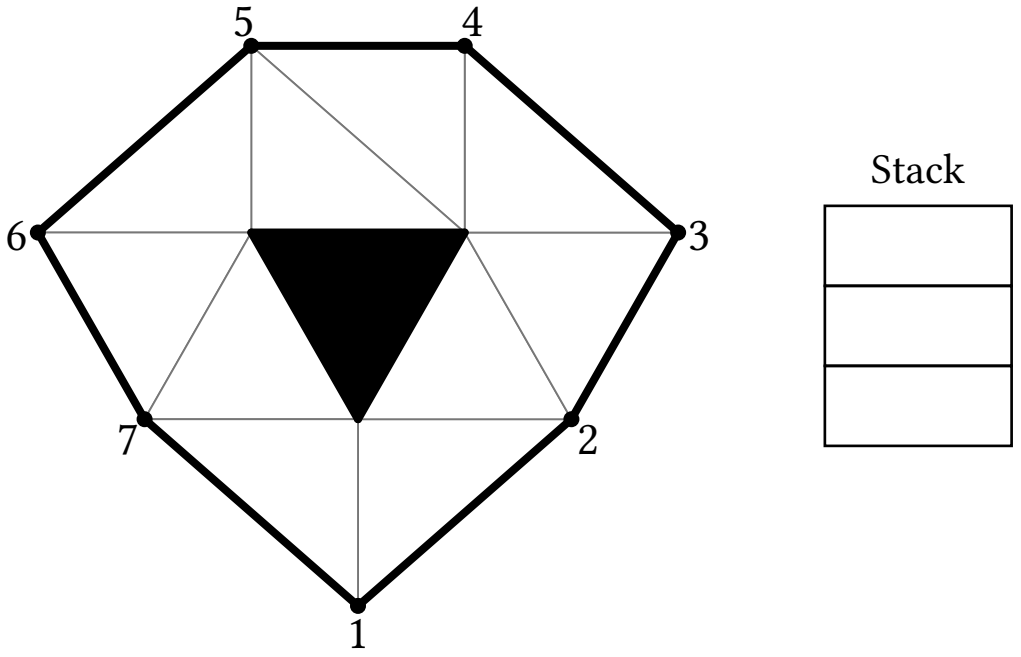


Figure 6: Komprese díry.

3.3 Vyšší genus?

History = S

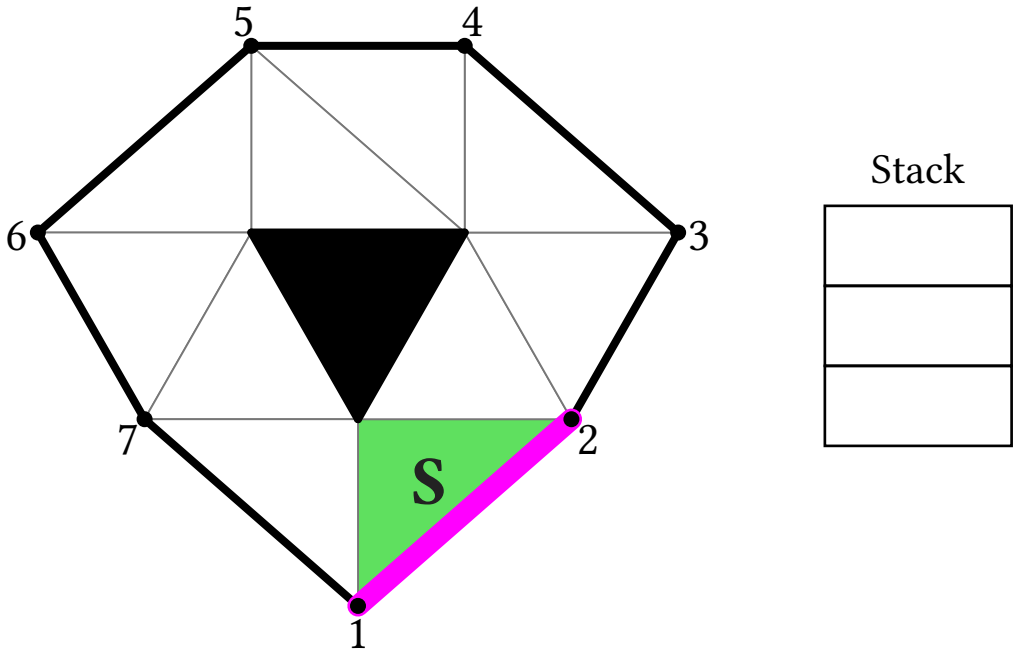
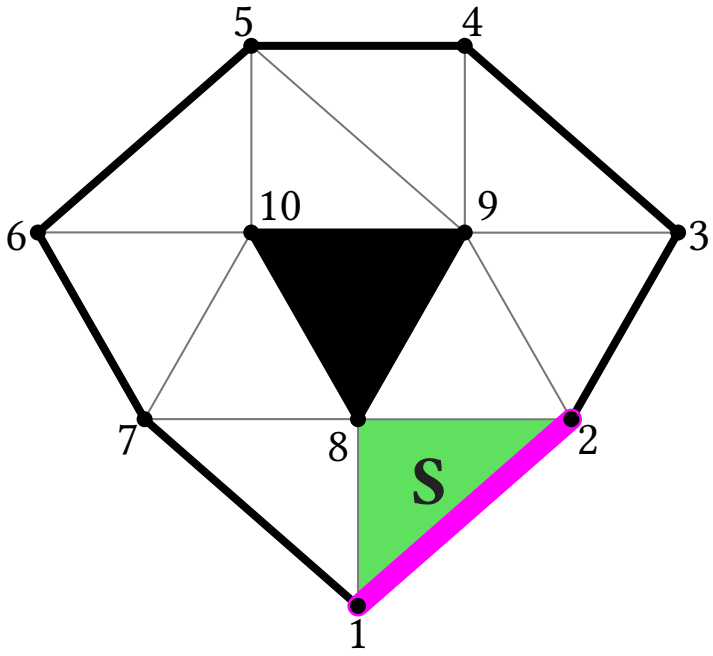


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = S



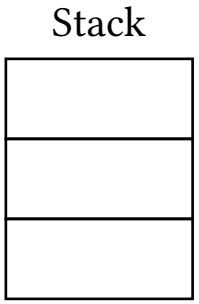


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SL

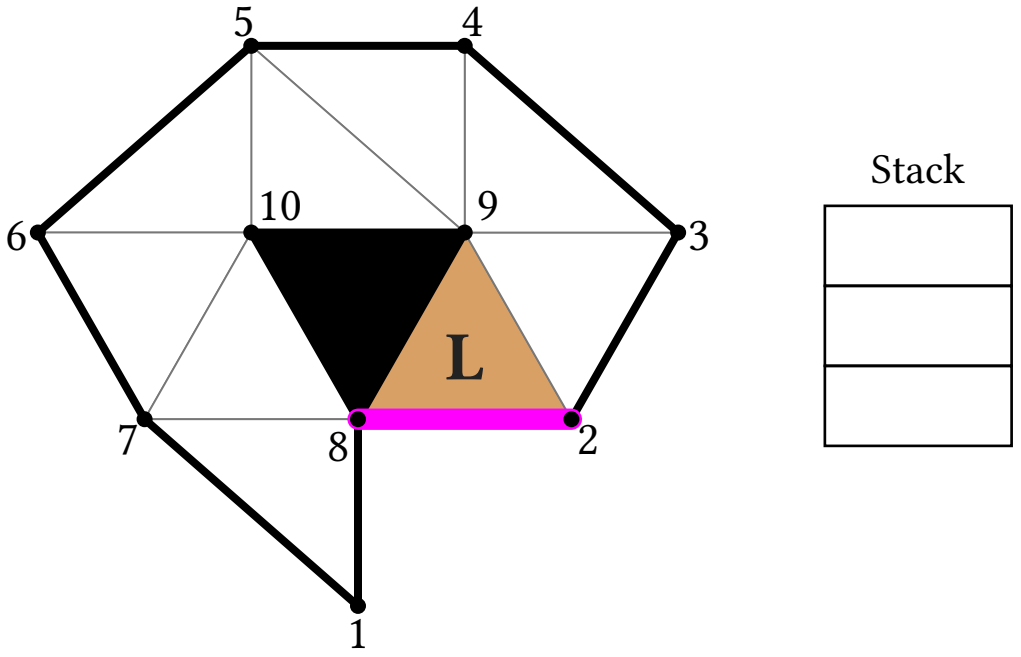


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLR

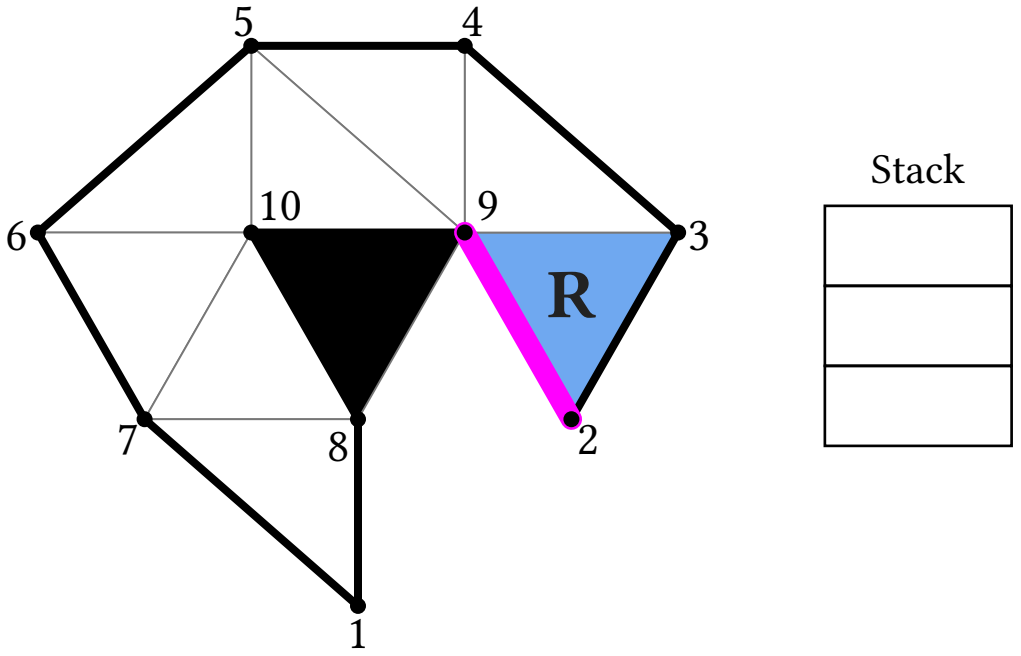


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLRR

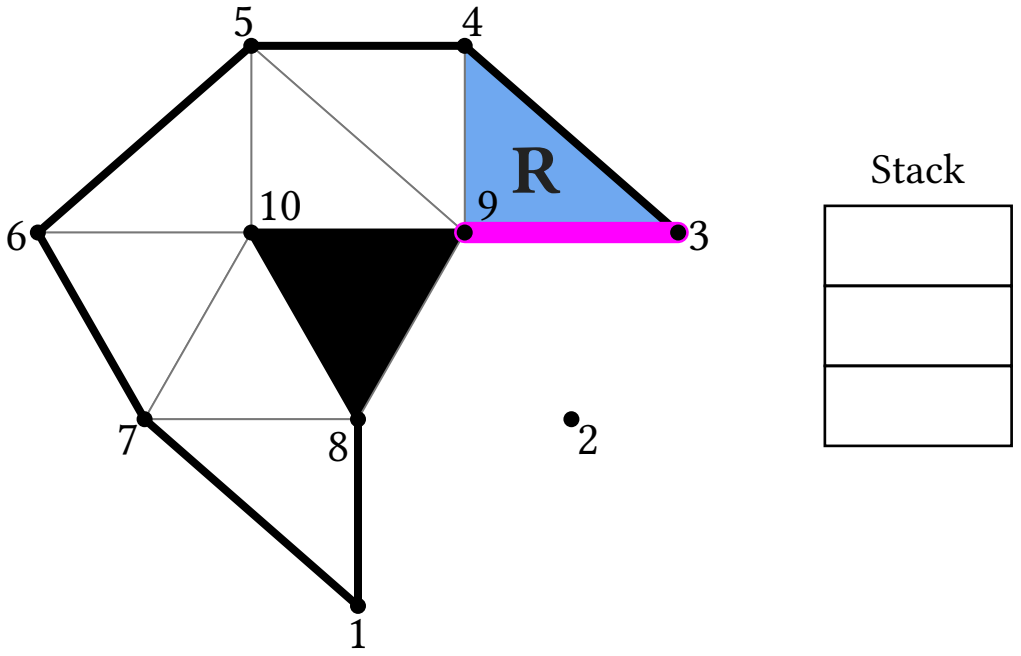


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLRRR

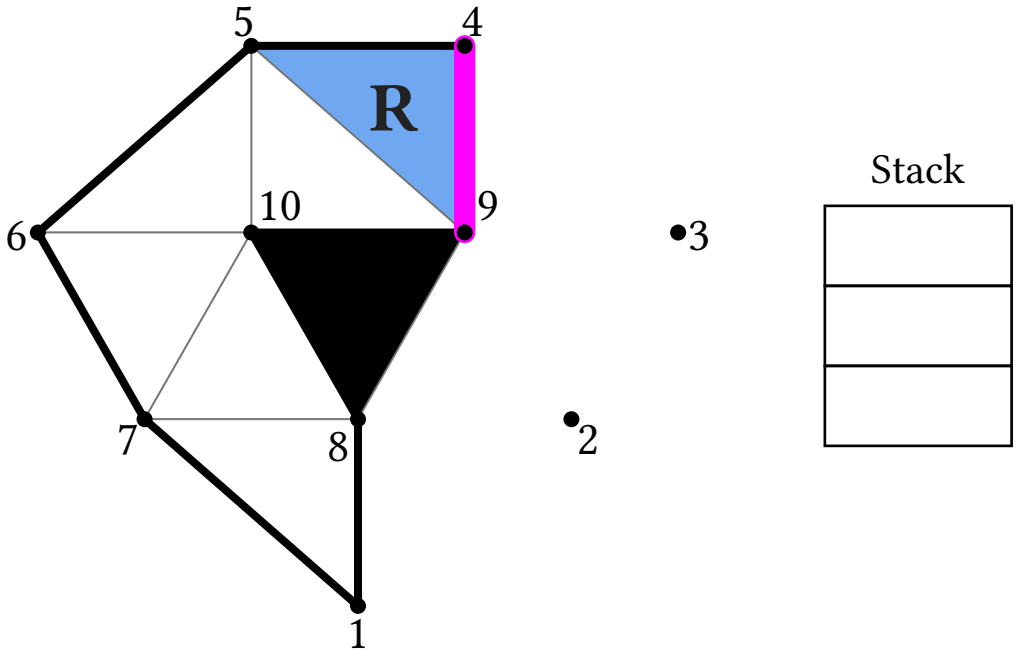


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLRRRL

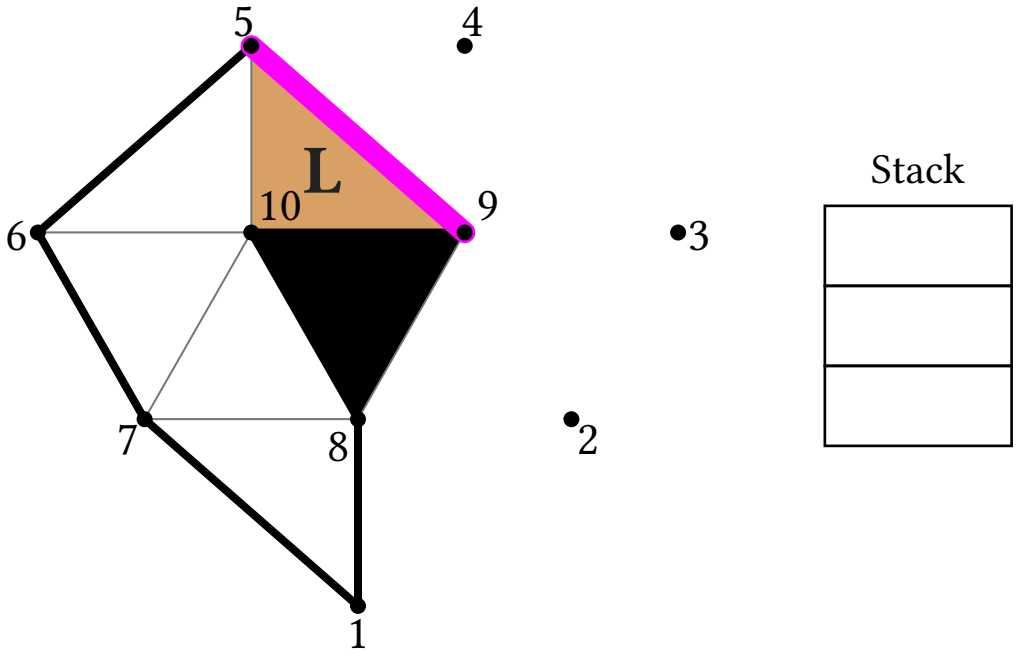


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLRRRLR

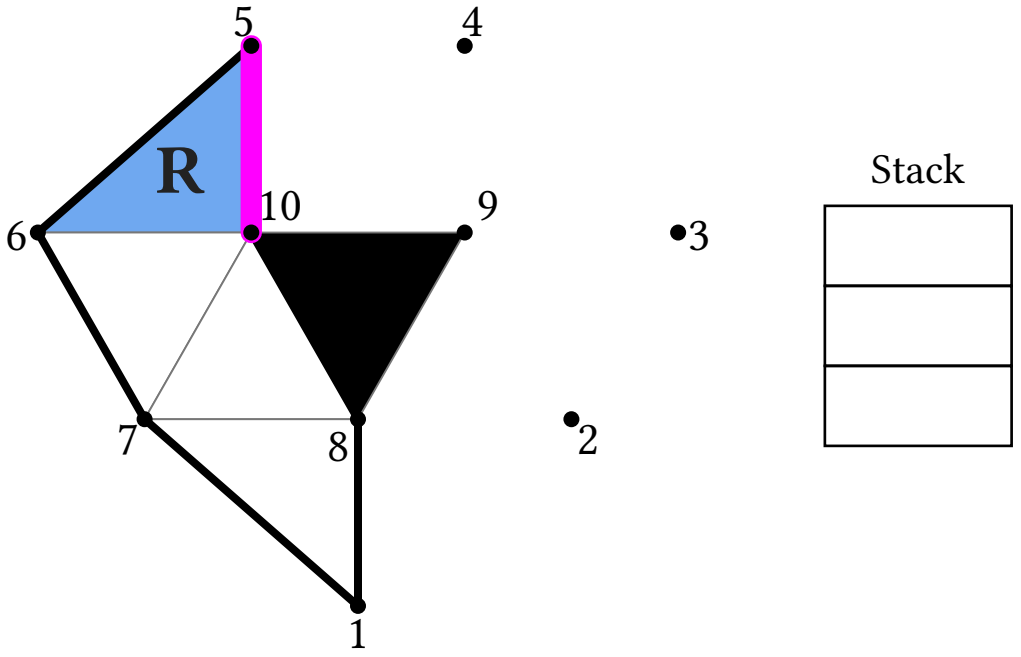


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLRRRLRR

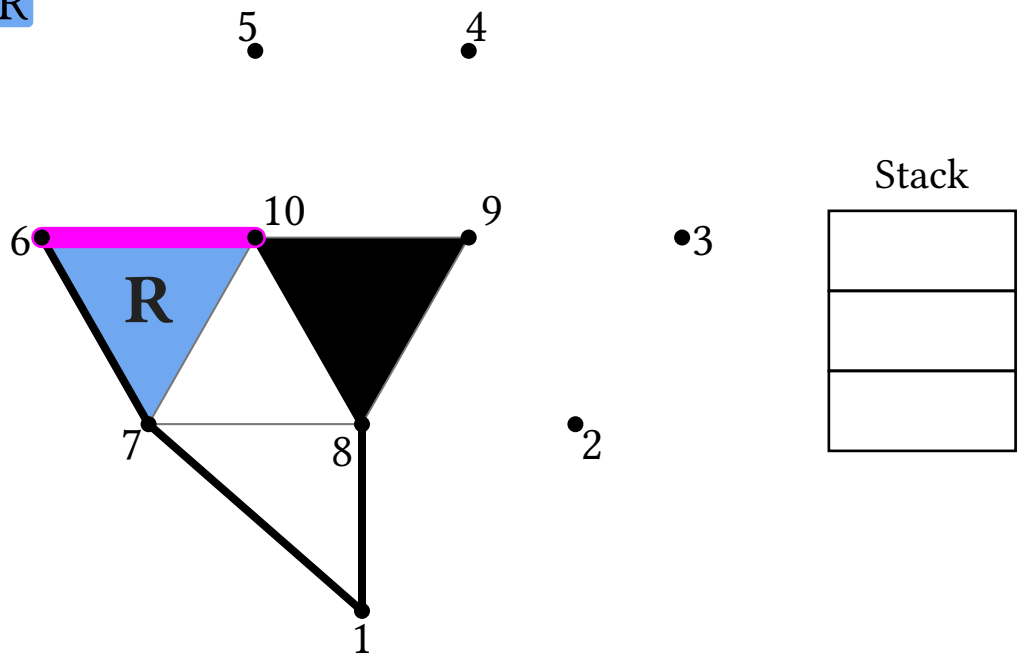


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLRRRLRRL

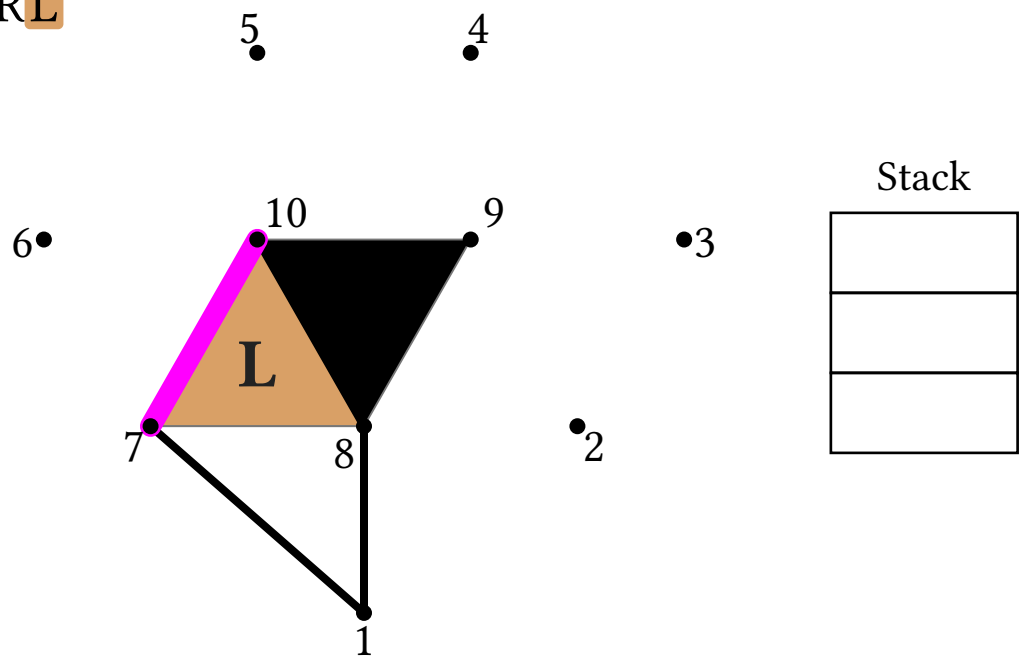


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLRRRLRRLE

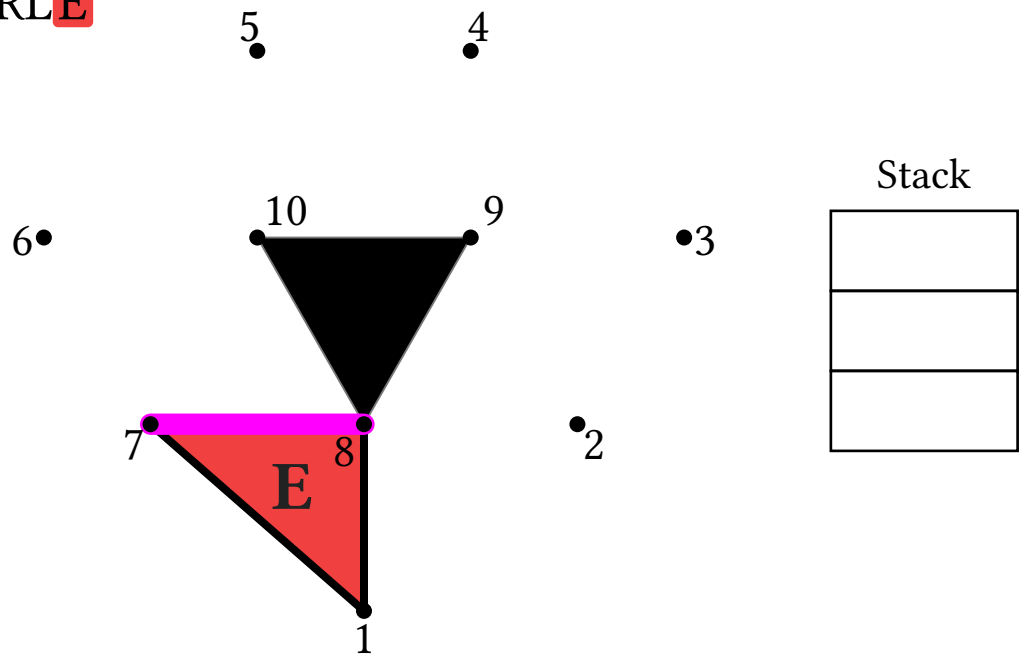


Figure 8: Komprese díry.

3.3 Vyšší genus?

History = SLRRRLRRLE

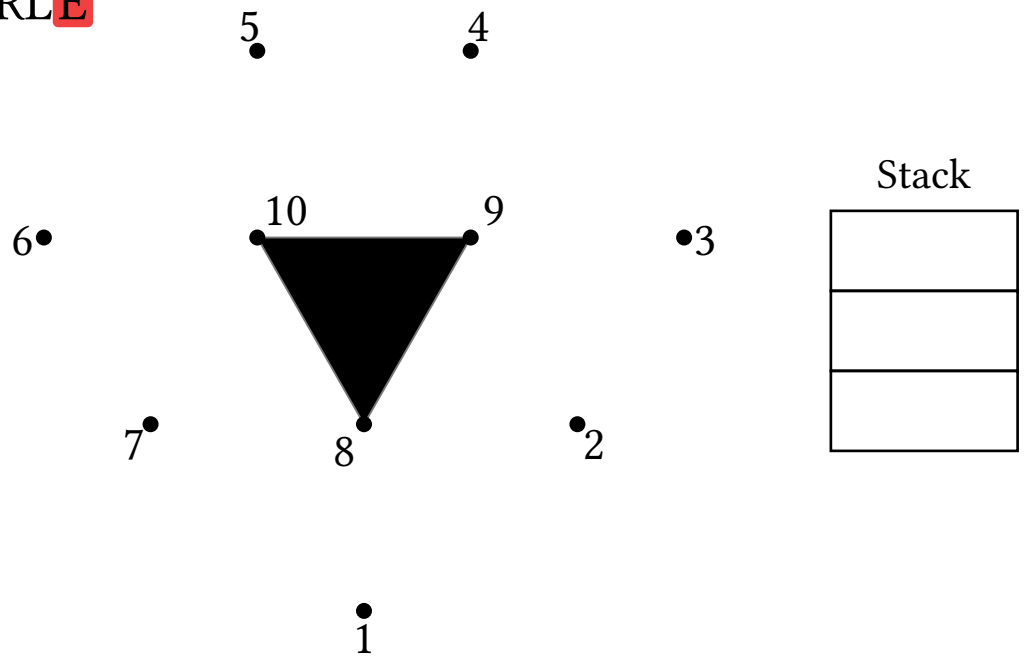


Figure 8: Komprese díry.

4. Dekomprese

4.1 Možnosti

4.2 Pravidla

History = CCRRRSLCRSERRELCRRRCRRRE

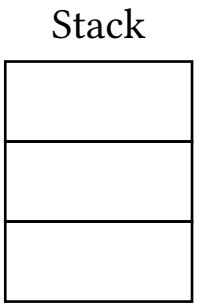


Figure 9: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRRE

Stack

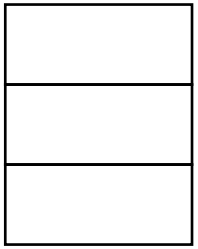


Figure 9: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRRE



Figure 9: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRR



Figure 9: Dekomprese.

5. Zdroje

5.1 Seznam figures

Figure 1: Dobový kontext vzniku algoritmu Edgebreaker. 2
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5.2 Obrázky (dobový kontext)

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3	**Windows 98** (5/1998)	SoundCloud (artwork)
4	**MrBeast** (5/1998)	YouTube (avatar kanálu)
5	**Google** (9/1998)	logodix.com
6	**Half-Life** (11/1998)	Valve, valvearchive.com Wikipedia *fair use*
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10	**Napster** (6/1999)	Jusore, GPLv3, Wikimedia Commons
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5.3 Reference

Originální paper [1] Corner table [2] Významné vylepšení [3] Wrap & Zip [4] Inspirace animací [5]

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